

The background of the slide is a dark, black sky filled with numerous small, glowing paper lanterns. These lanterns are scattered across the frame, some appearing closer and larger, while others are further away and smaller. They are all lit from within, casting a warm, orange-yellow glow. The overall effect is a soft, bokeh-like pattern of light against the dark background.

Permanent Maxillary Central Incisors

Dr Saba Arshad

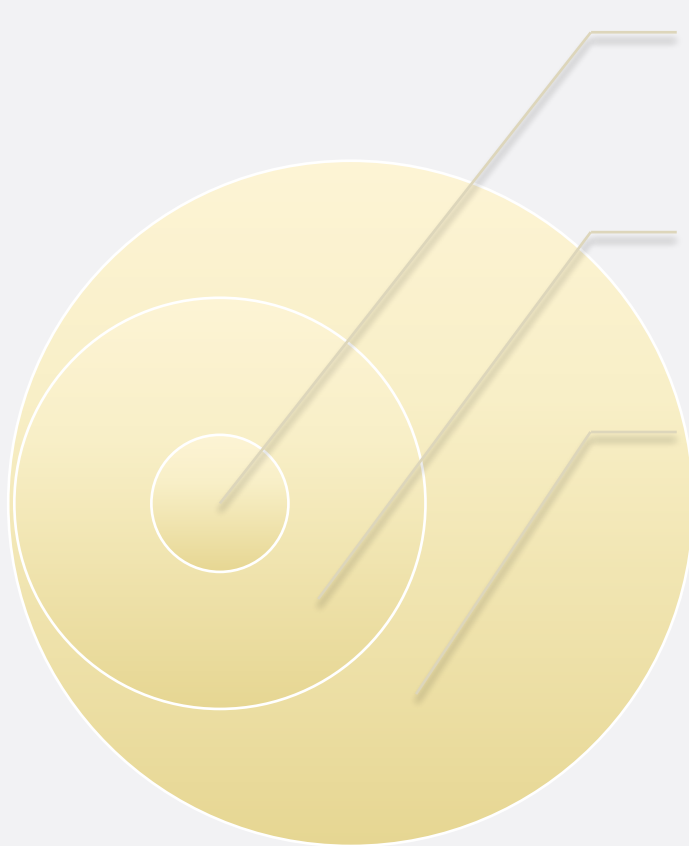
Senior Lecturer

Department of Oral Biology

Learning Objectives

- To define and pronounce the terminologies of permanent dentition.
- To label the anatomical landmarks of permanent maxillary central incisor.
- To discuss the curves and segments of maxillary central incisors.
- To be able to correctly draw the graph outline.
- To be able to carve the teeth on wax models.
- To discuss anatomical variations.

Why it is important to describe every curve and segment of normal tooth



To understand the functional basis.

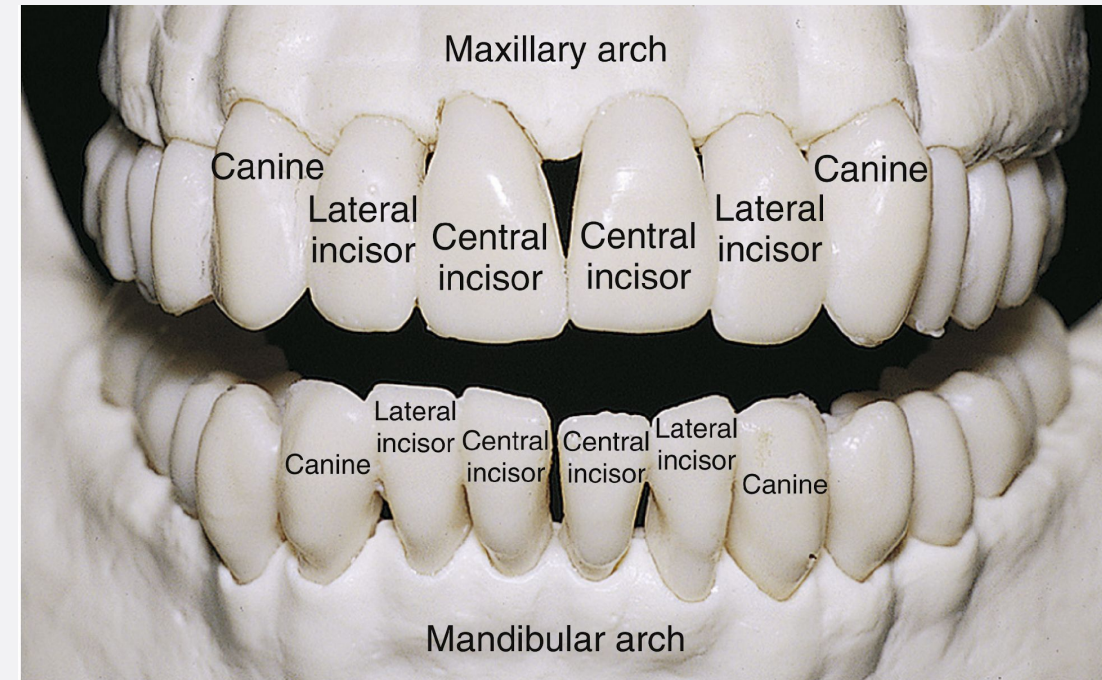
To establish arbitrary norm for individual teeth.

- **For comparative study of natural sizes, their normal alignment and occlusion.**
- **To reproduce the anatomy accurately to restore the defected or lost tooth form**

MAXILLARY CENRAL INCISOR

The right and left maxillary central incisors lie along the midline of the oral cavity.

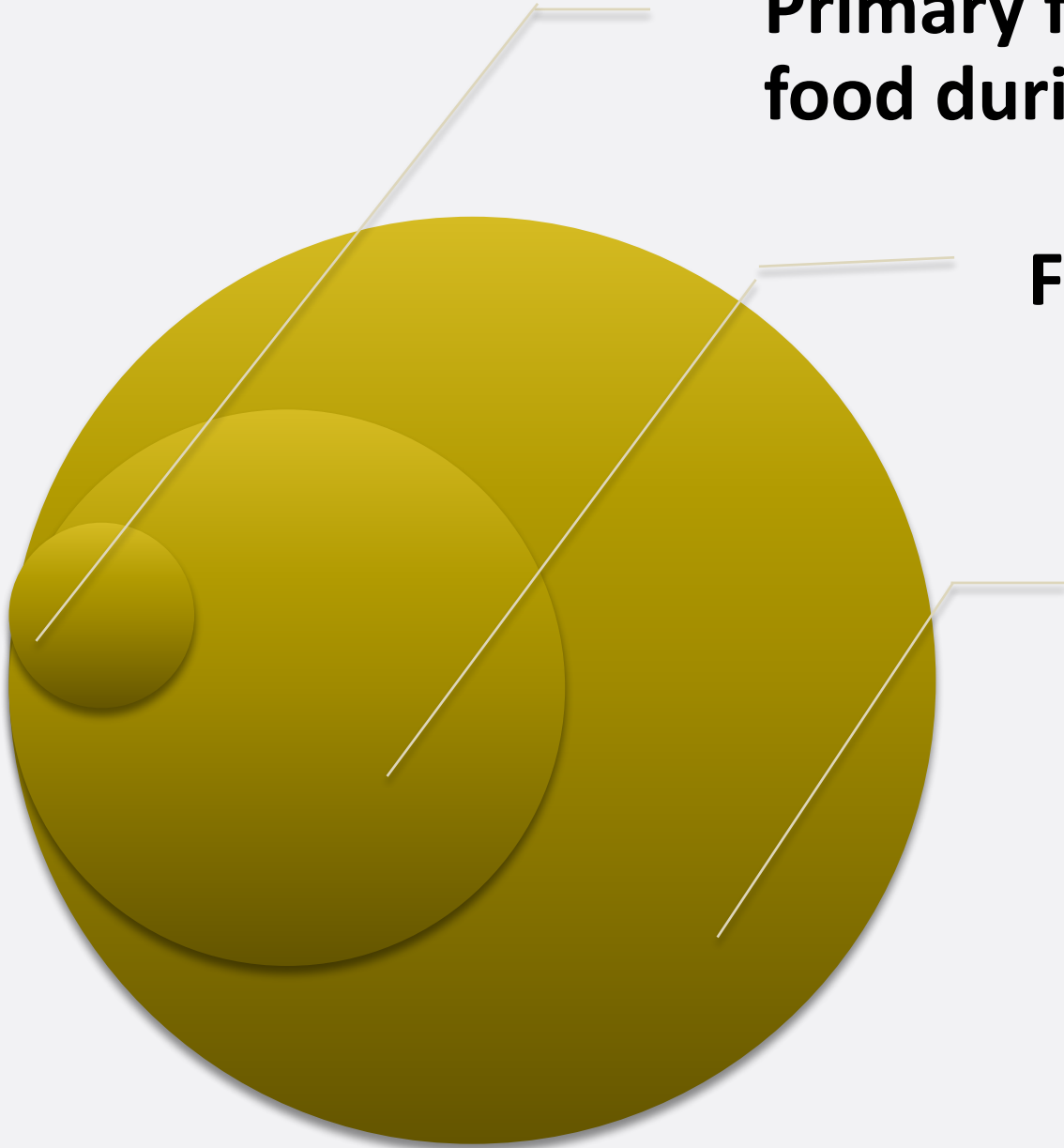
The maxillary lateral incisor is smaller in all dimensions except root length than the central incisor.



**Primary function is shearing and cutting
food during mastication**

Facial aesthetics

Phonetics



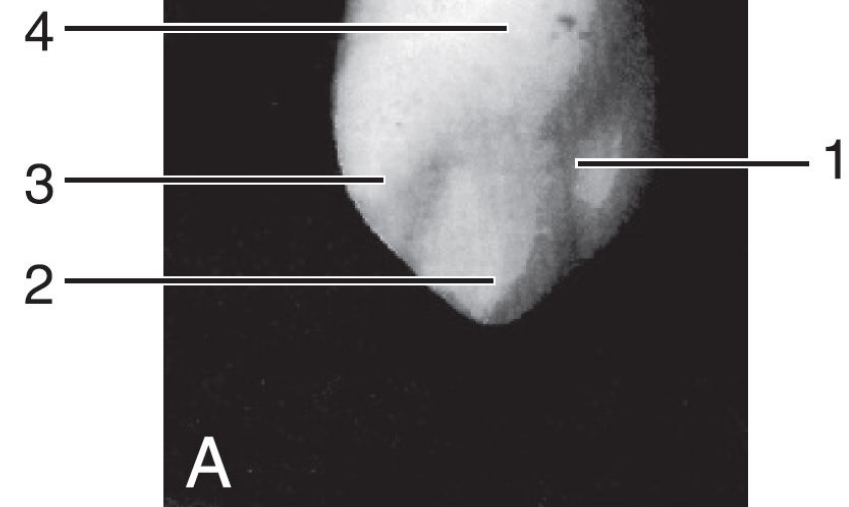
Developmental Lobes

- **A, Labial aspect of maxillary central incisor, showing the labial grooves marking the division of the lobes.**
- **1, Mesial lobe;**
- **2, labial lobe;**
- **3, distal lobe.**
- **4, The lingual lobe, or Cingulum,**



FIGURE 4-12 General outlines of some of the lobes.

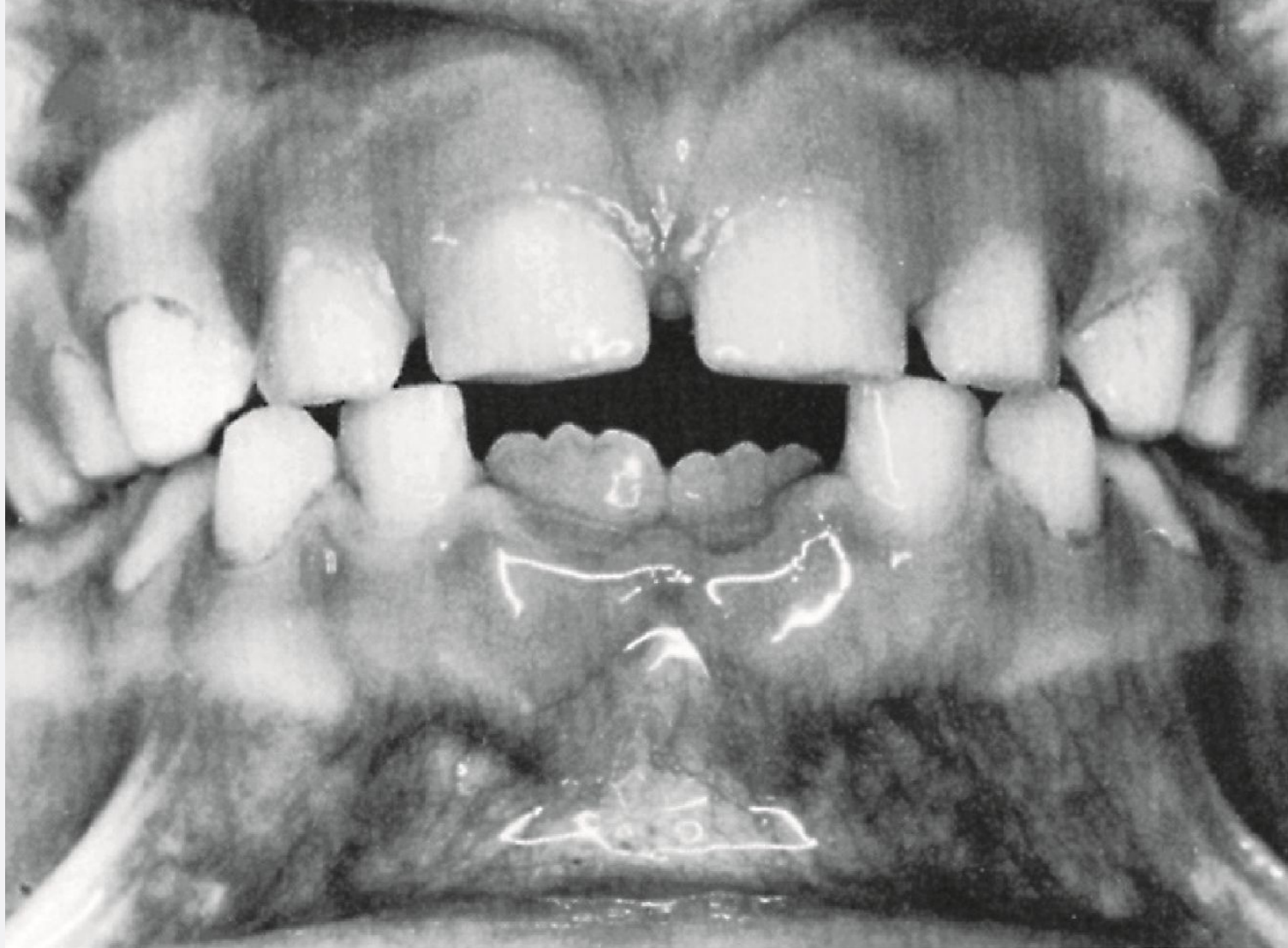
Identify the
developmental
lobes of
maxillary
central incisors



Chronology of Permanent Maxillary Incisors

1st Evidence Of Calcification	3–4 Months
Enamel (Crown) Completed	4–5 Years
Eruption	7–8 Years
Root Completed	10 Years

6-7 years old



7-8 years old

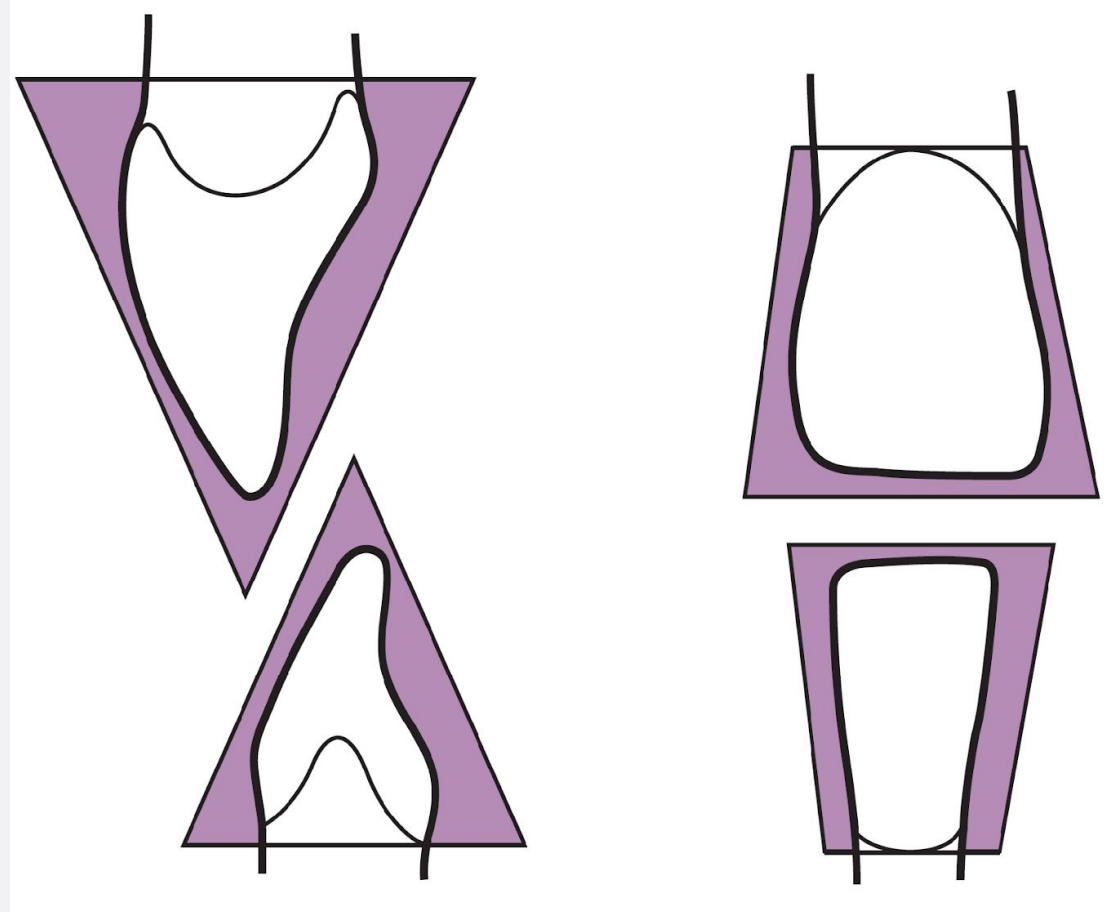


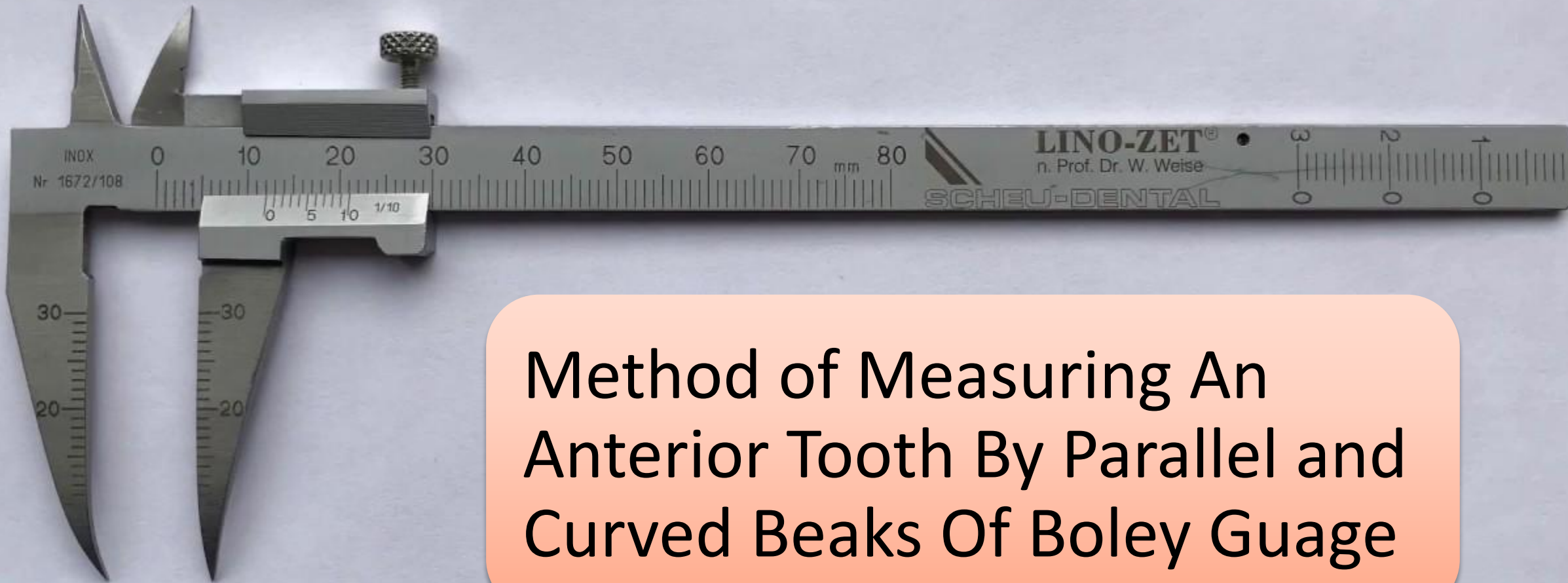
Erupting #7 and #8

7-8 years old



Geometric Outline of Surfaces



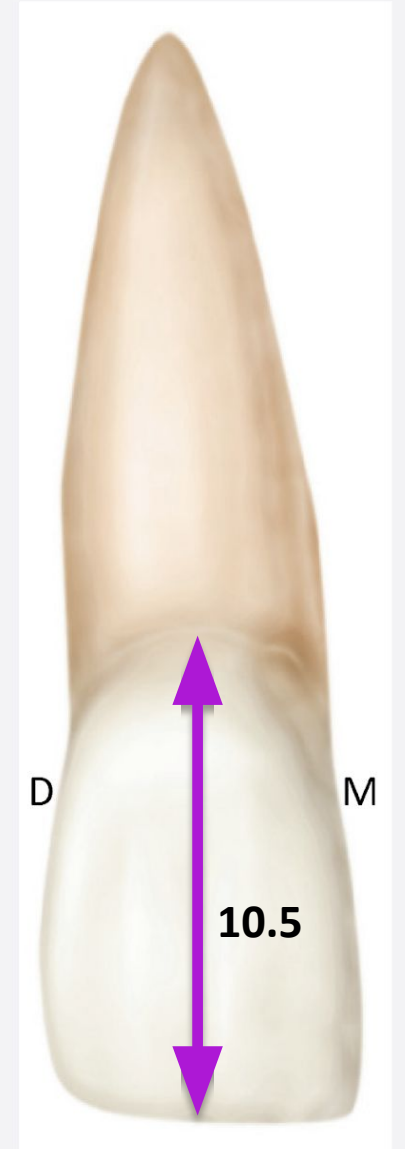
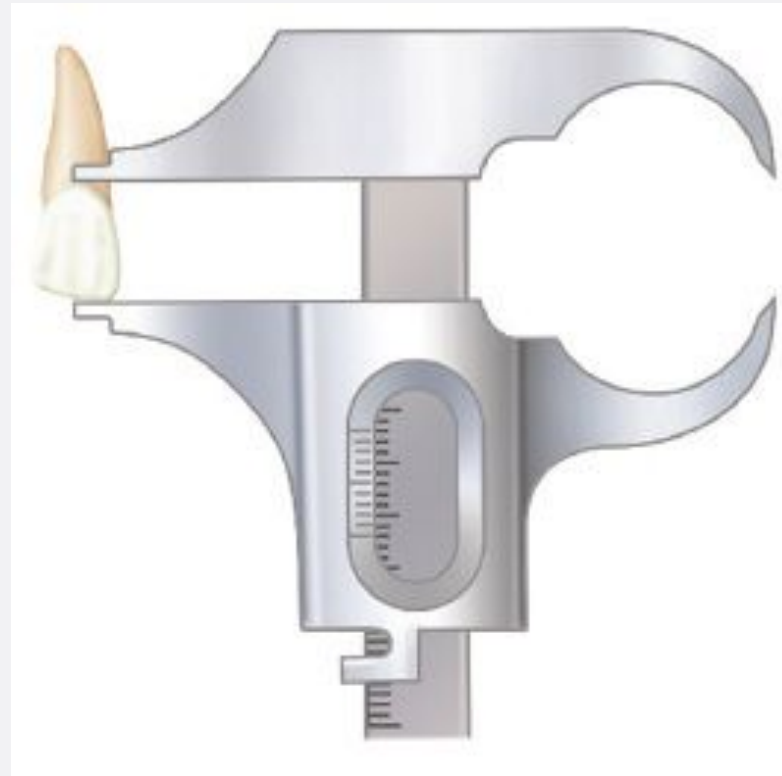


Method of Measuring An Anterior Tooth By Parallel and Curved Beaks Of Boley Guage

Dimensions of Maxillary Central Incisor

**Cervico-incisal
Length of Crown**

10.5mm

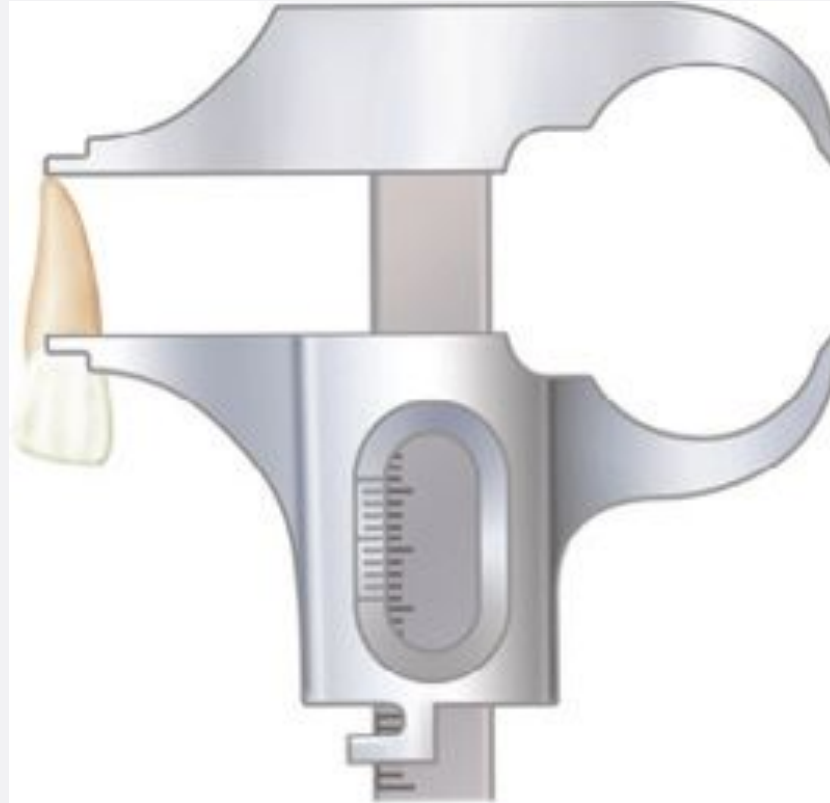


Maxillary central incisor

Dimensions of Maxillary Central Incisor

Length of Root

13mm

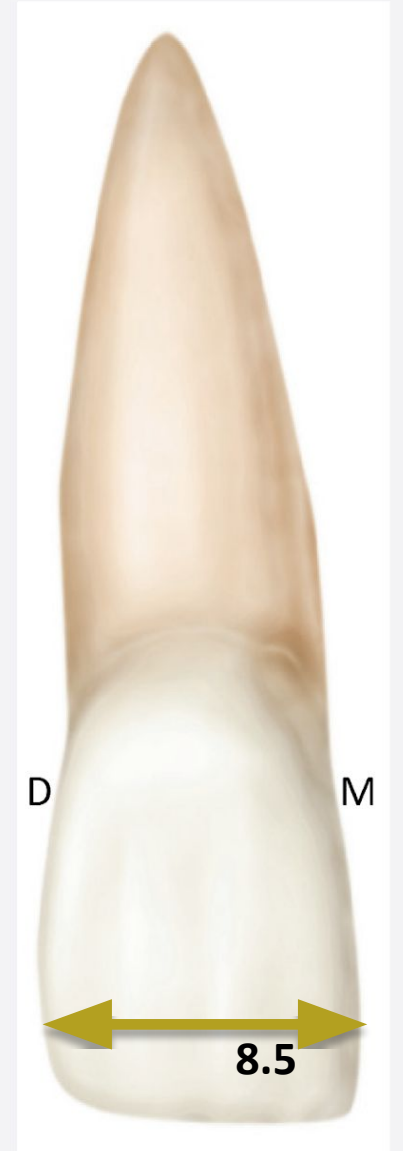
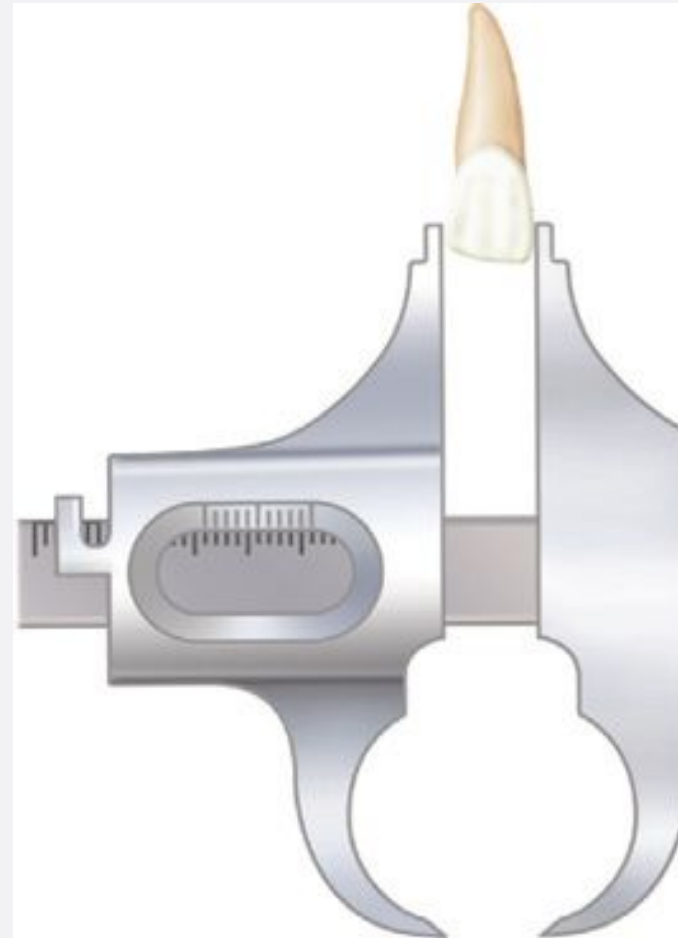


Maxillary central incisor

Dimensions of Maxillary Central Incisor

**Mesiodistal
Diameter of
Crown**

8.5mm



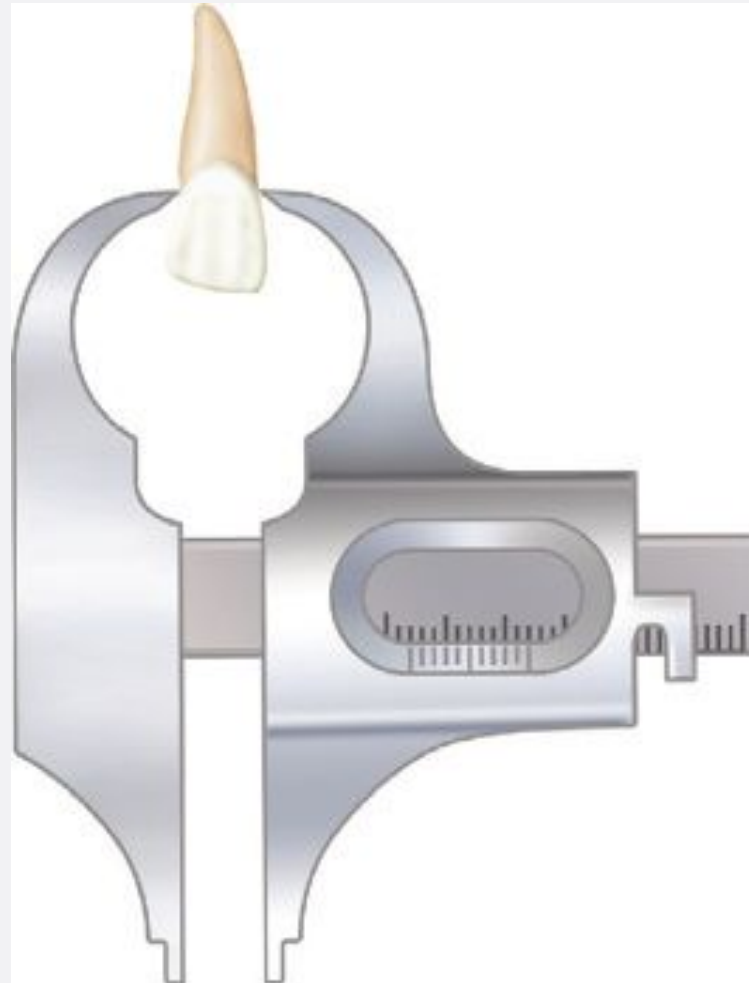
Maxillary central incisor

Dimensions of Maxillary Central Incisor

**Mesiodistal Diameter of
Crown at Cervix**

7mm

**1.5 to 2mm less at cervical
line than incisal edge**

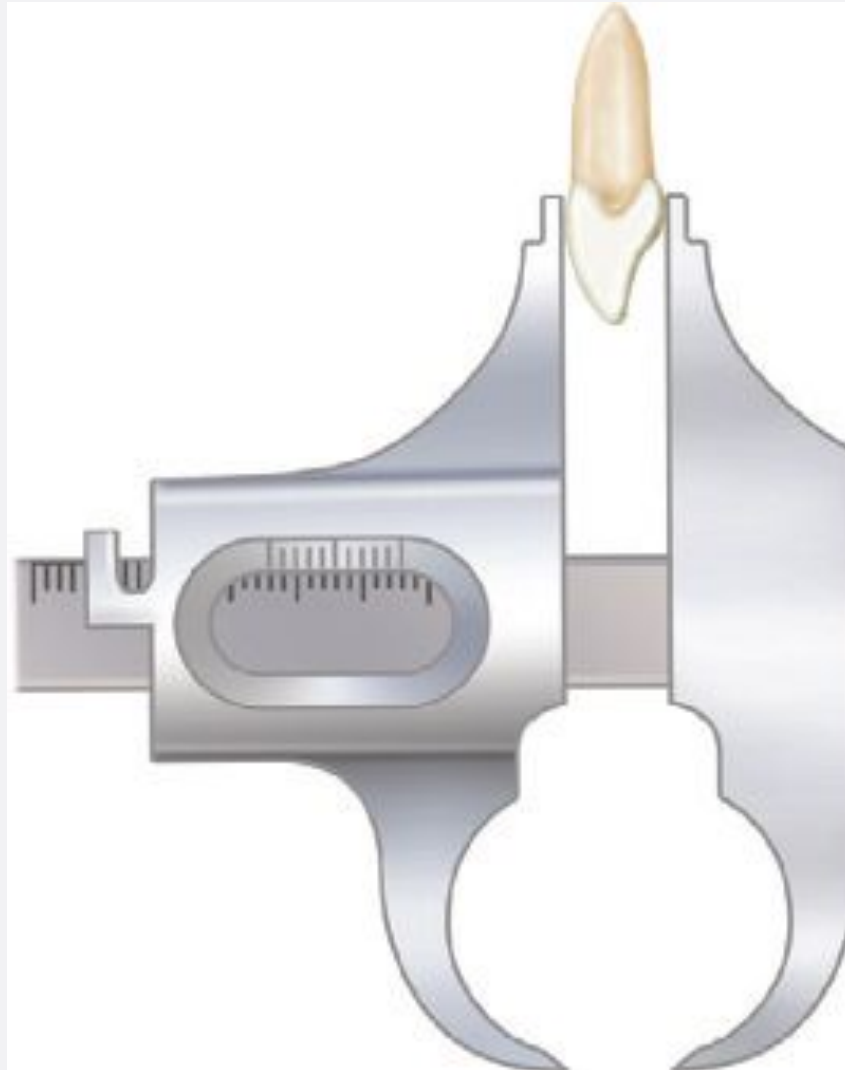


Maxillary central incisor

Dimensions of Maxillary Central Incisor

**Labio-lingual
Diameter of
Crown**

7mm



Mesial

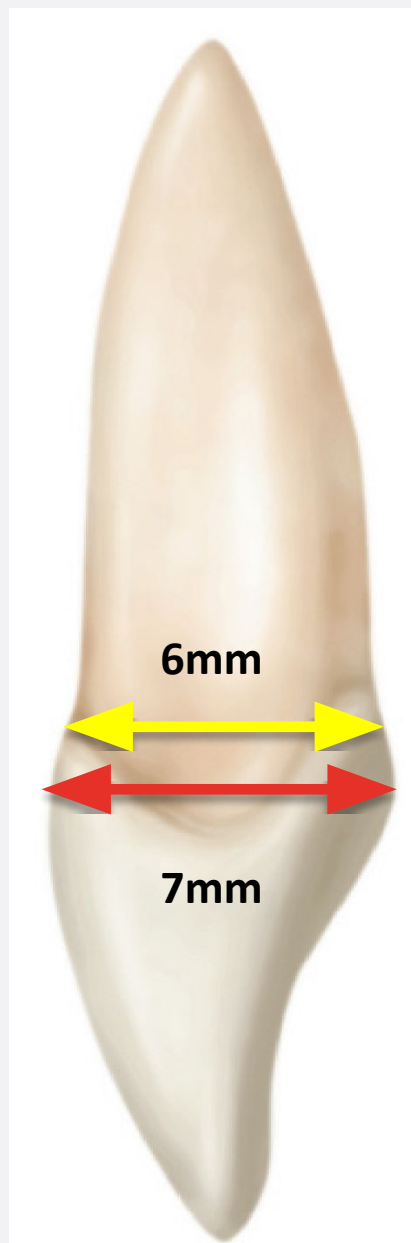
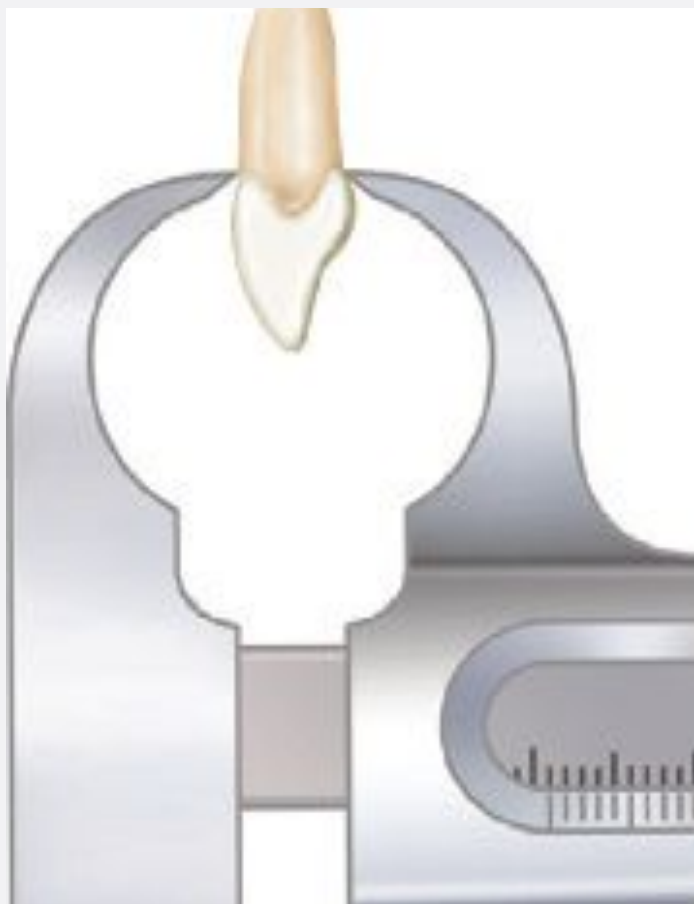


Distal

Dimensions of Maxillary Central Incisor

**Labio-lingual Diameter Of
Crown at Cervix**

6mm



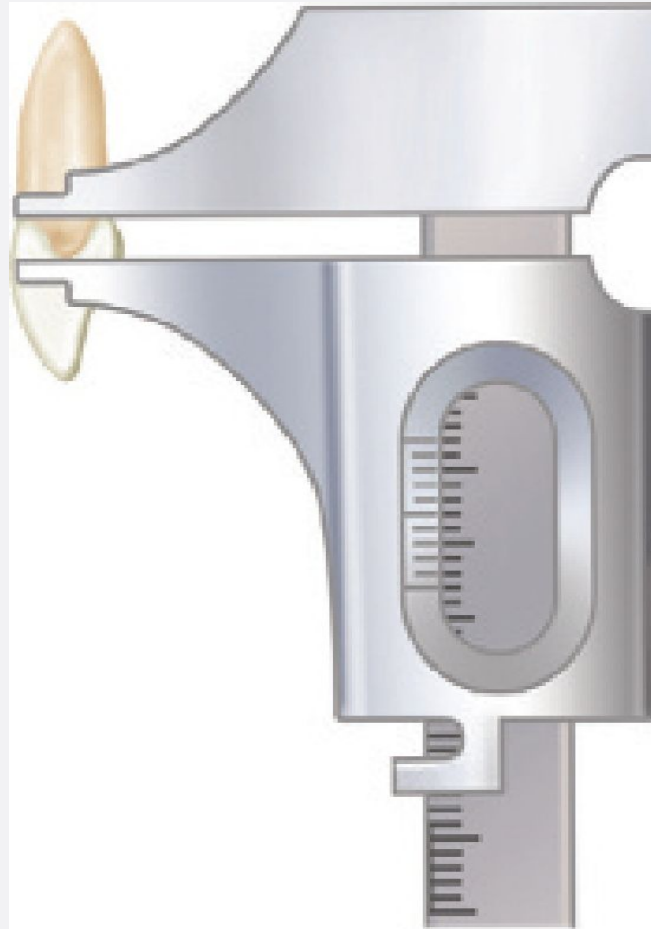
Mesial



Distal

Dimensions of Maxillary Central Incisor

Curvature of Cervical Line— Mesial	Curvature Of Cervical Line— Distal
3.5mm	2.5mm

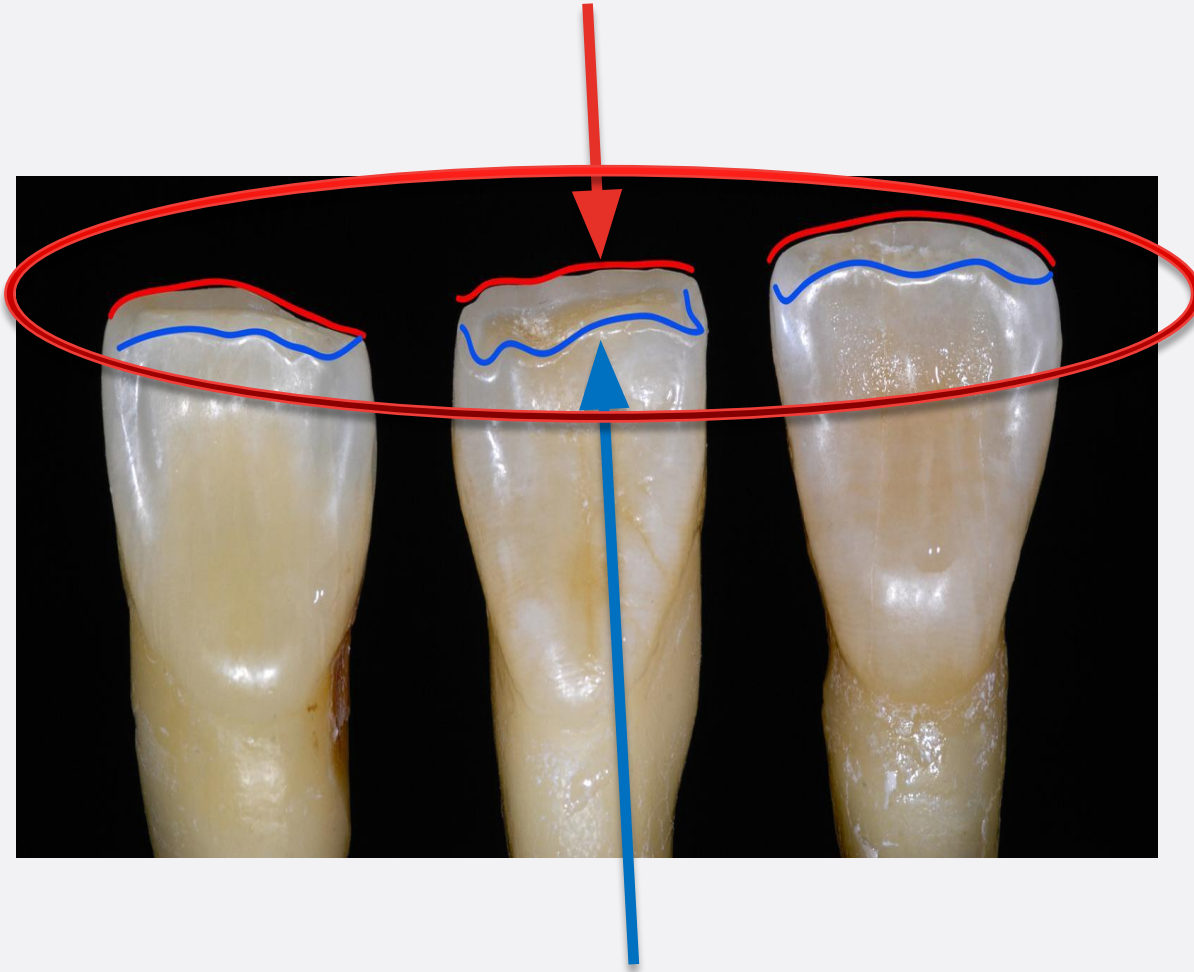


Mesial



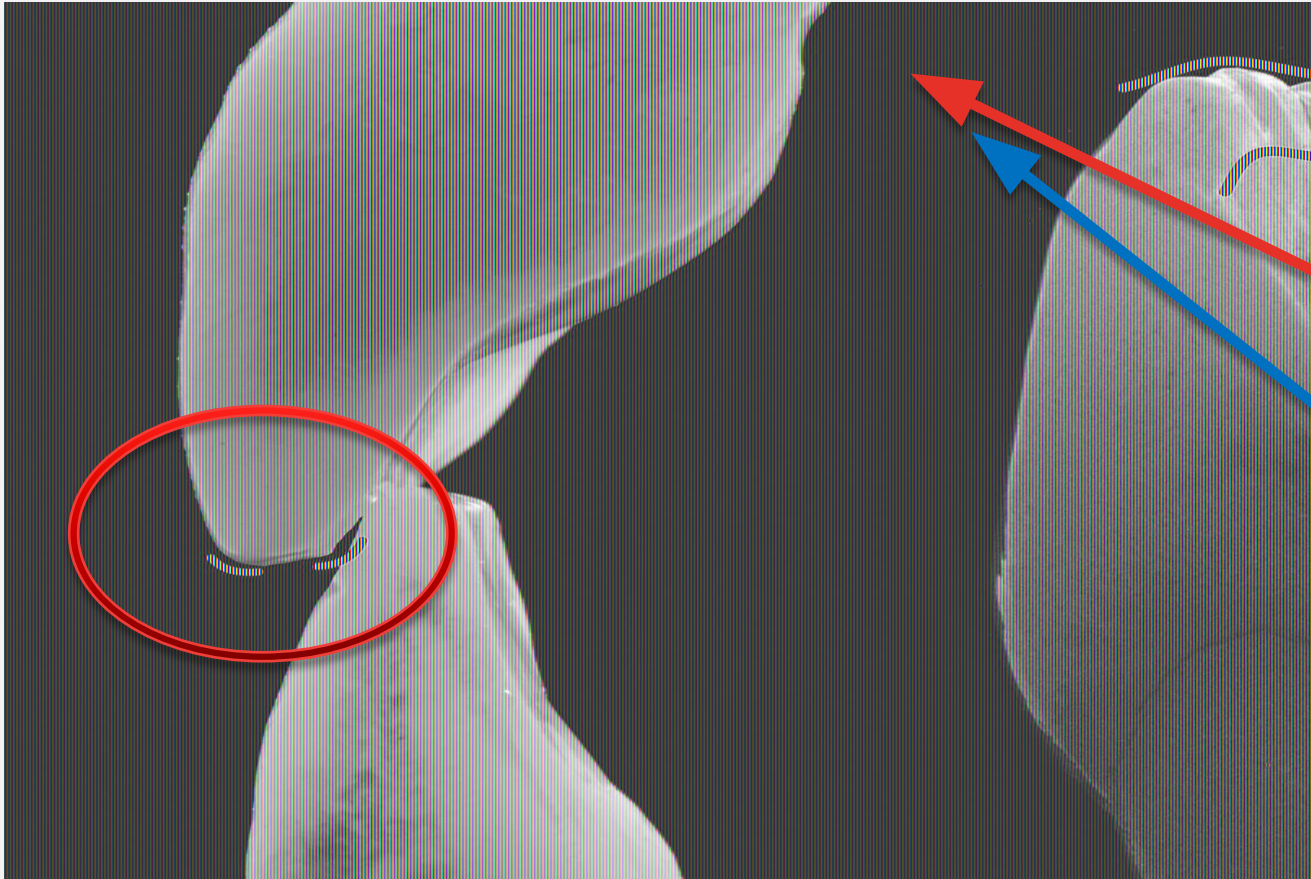
Distal

Incisal Ridge



- It is the complete incisal portion of the crown that is rounded and merged with the mesio-incisal and disto-incisal angles and the labial and lingual surfaces in newly erupted incisors.
- After occlusal wear, it splits into two components;
- **labio-incisal edge (LIE)**
- **linguo-incisal edge (LIE)**

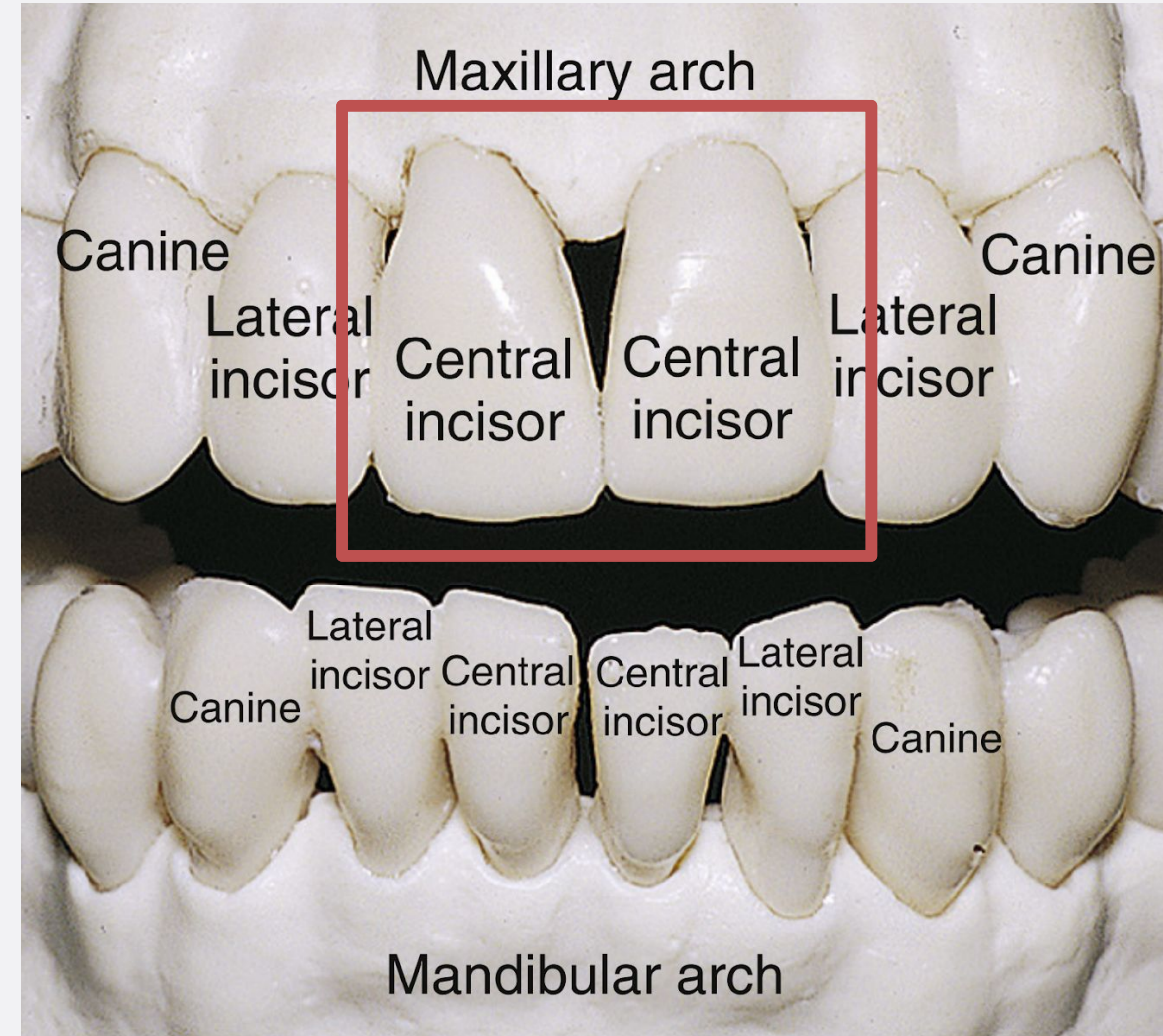
Incisal Edge



- Due to occluding and masticatory stresses
- The **incisal edge** that is an angle formed by merging of two flattened surfaces;
 1. linguo-incisal surface
AKA incisal surface
 2. labial surface.
- Mamelons lie on incisal edge and ridge

General Features

- **Widest mesiodistally among anterior**
- **Squared or rectangular appearance**
- **Symmetrical**
- **Nearly straight incisal edge**
- **Curvature of cervical line towards root**
- **Labial surface convex at cervical 3rd, flat at middle and incisal 3rd**

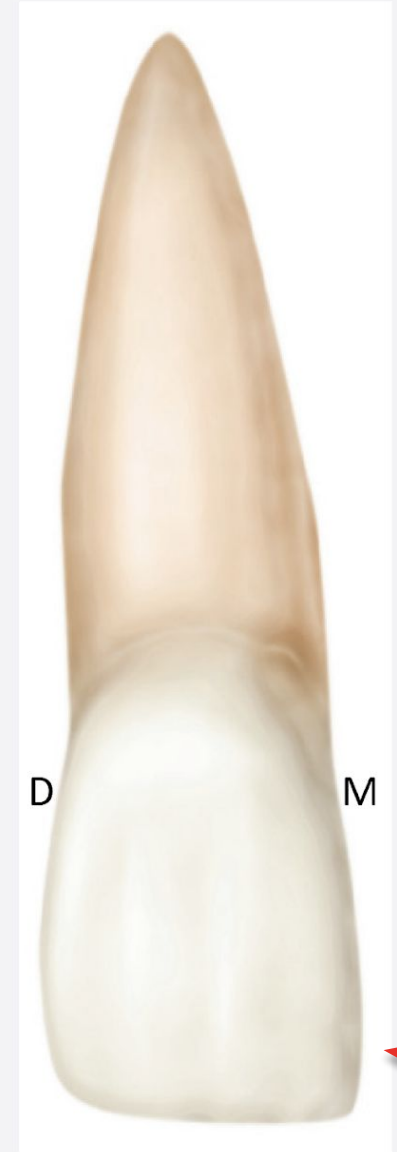
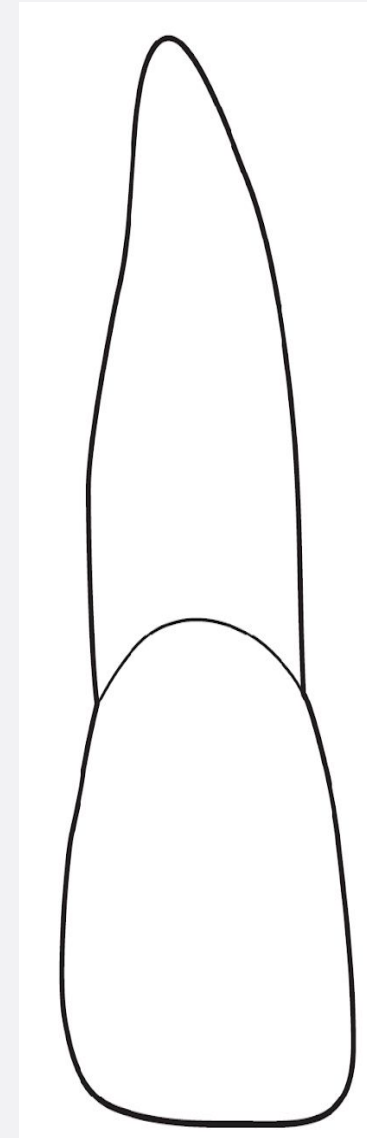
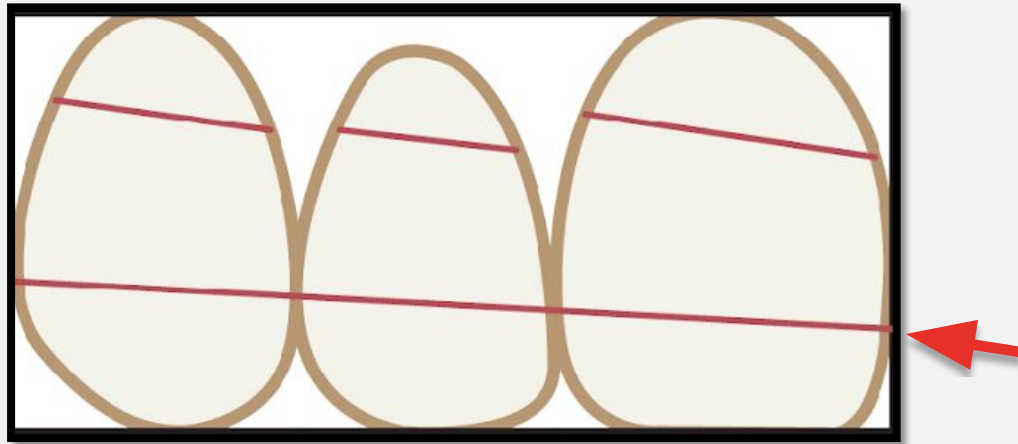


Labial surface

The mesial outline of the crown is only slightly convex,

with the crest of curvature at the incisal third of the crowns (representing the contact area)

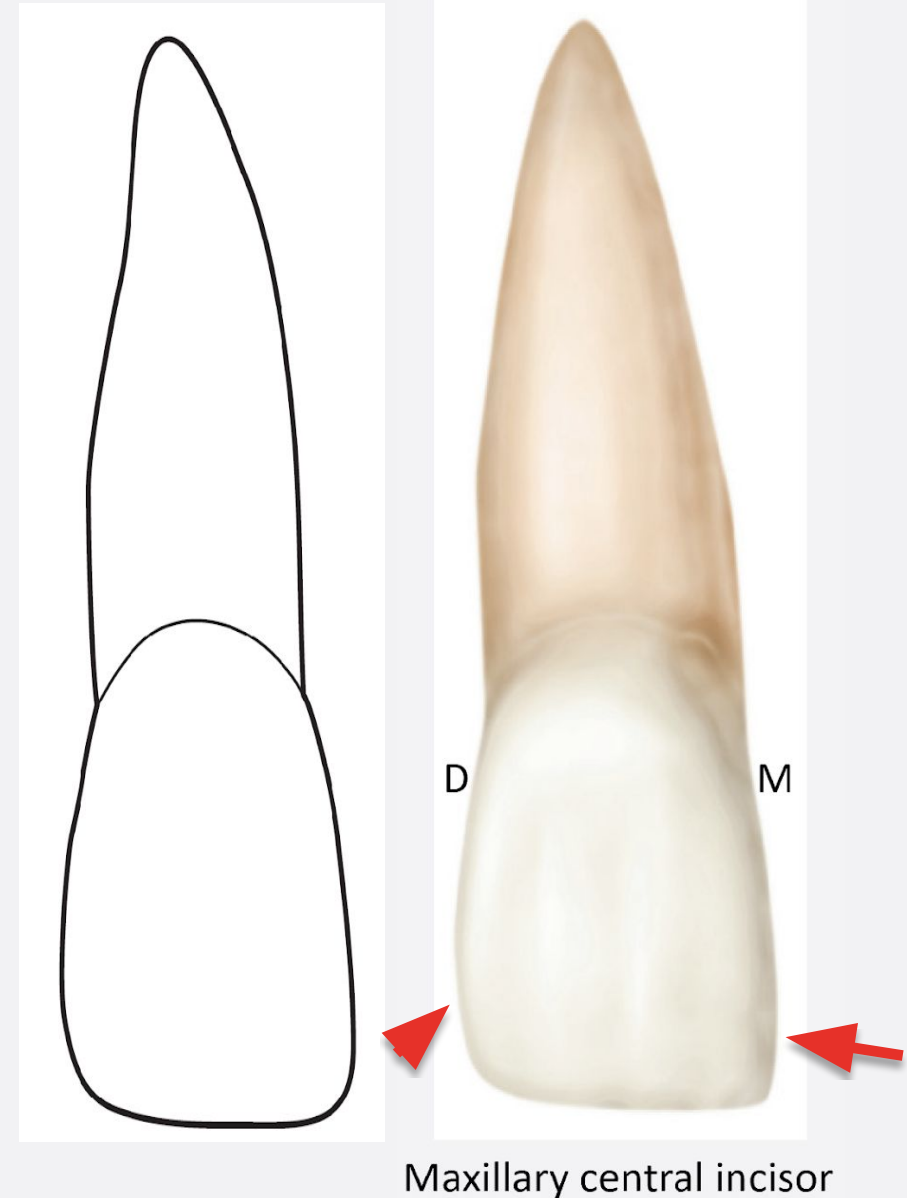
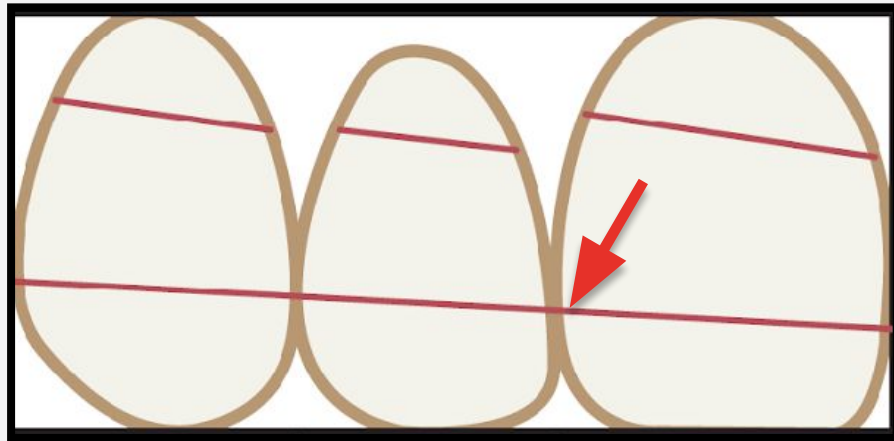
Approaching the mesio-incisal angle.



Maxillary central incisor

Labial Surface

The distal outline of the crown is more convex than mesial outline, with the crest of curvature higher toward the cervical line.



Labial Surface

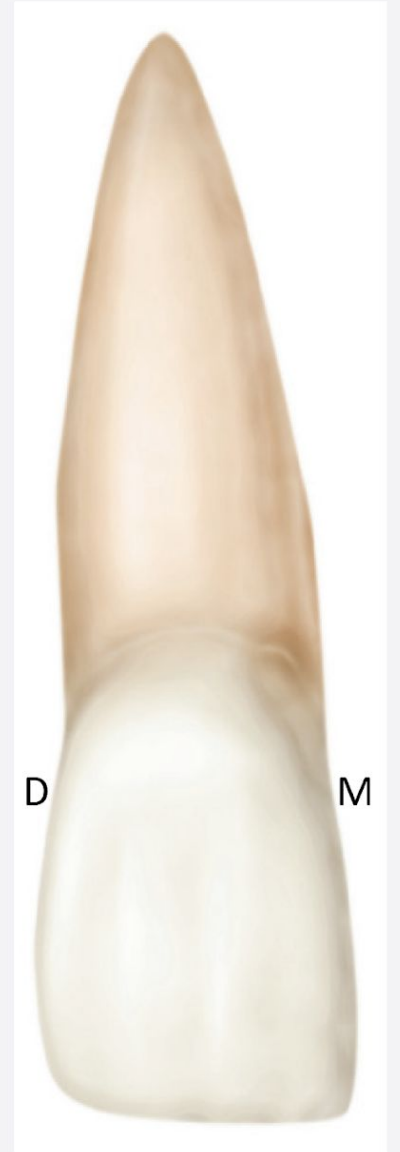
The disto-incisal angle is not as sharp as the mesio-incisal angle



Maxillary central incisor

Labial Surface

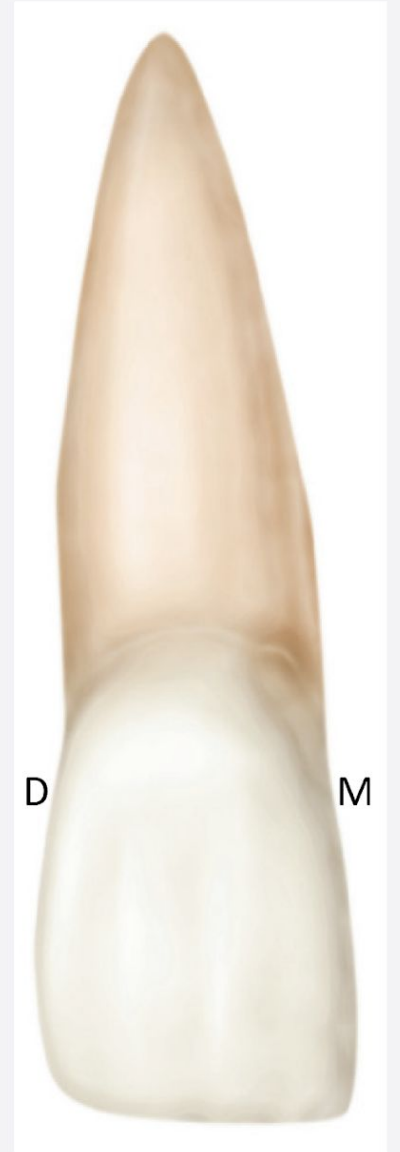
The incisal outline is usually regular and straight in a mesiodistal direction



Maxillary central incisor

Labial Surface

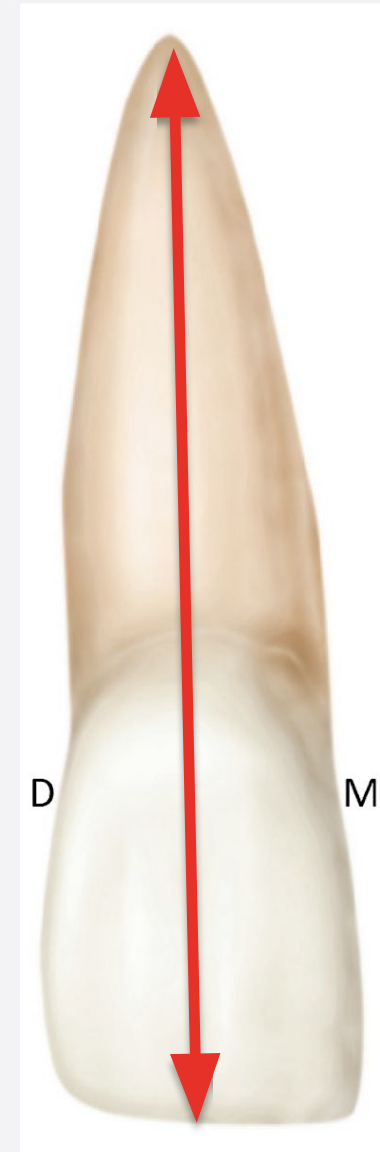
The cervical outline of the crown follows a semicircular direction with the curvature rootwise



Maxillary central incisor

Labial Surface

- The root is cone-shaped,
- with a relatively blunt apex and regular mesial and distal outline
- 2 or 3 mm longer than the crown



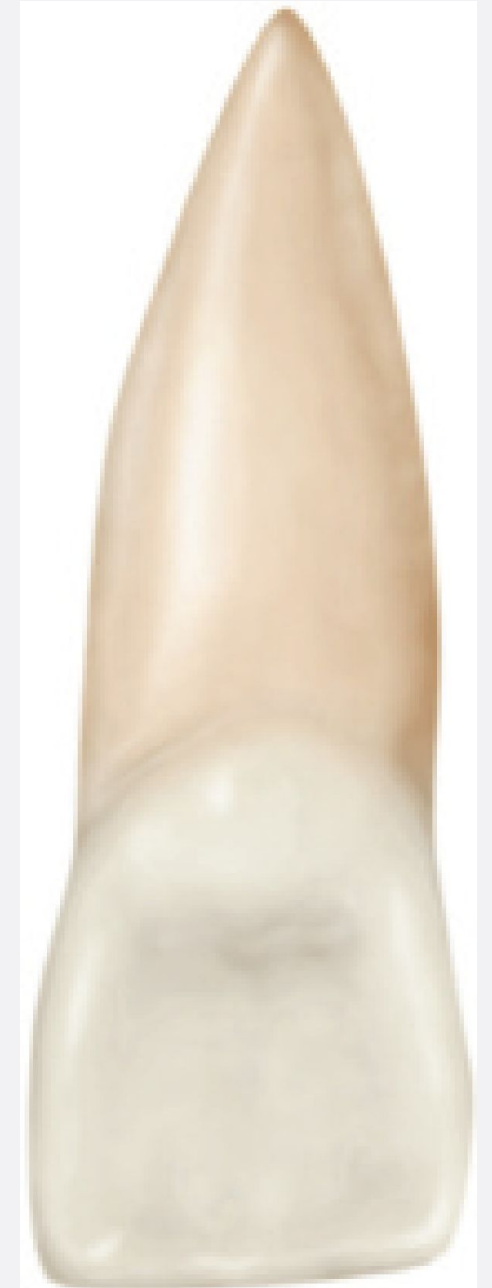
Maxillary central incisor

Lingual Surface



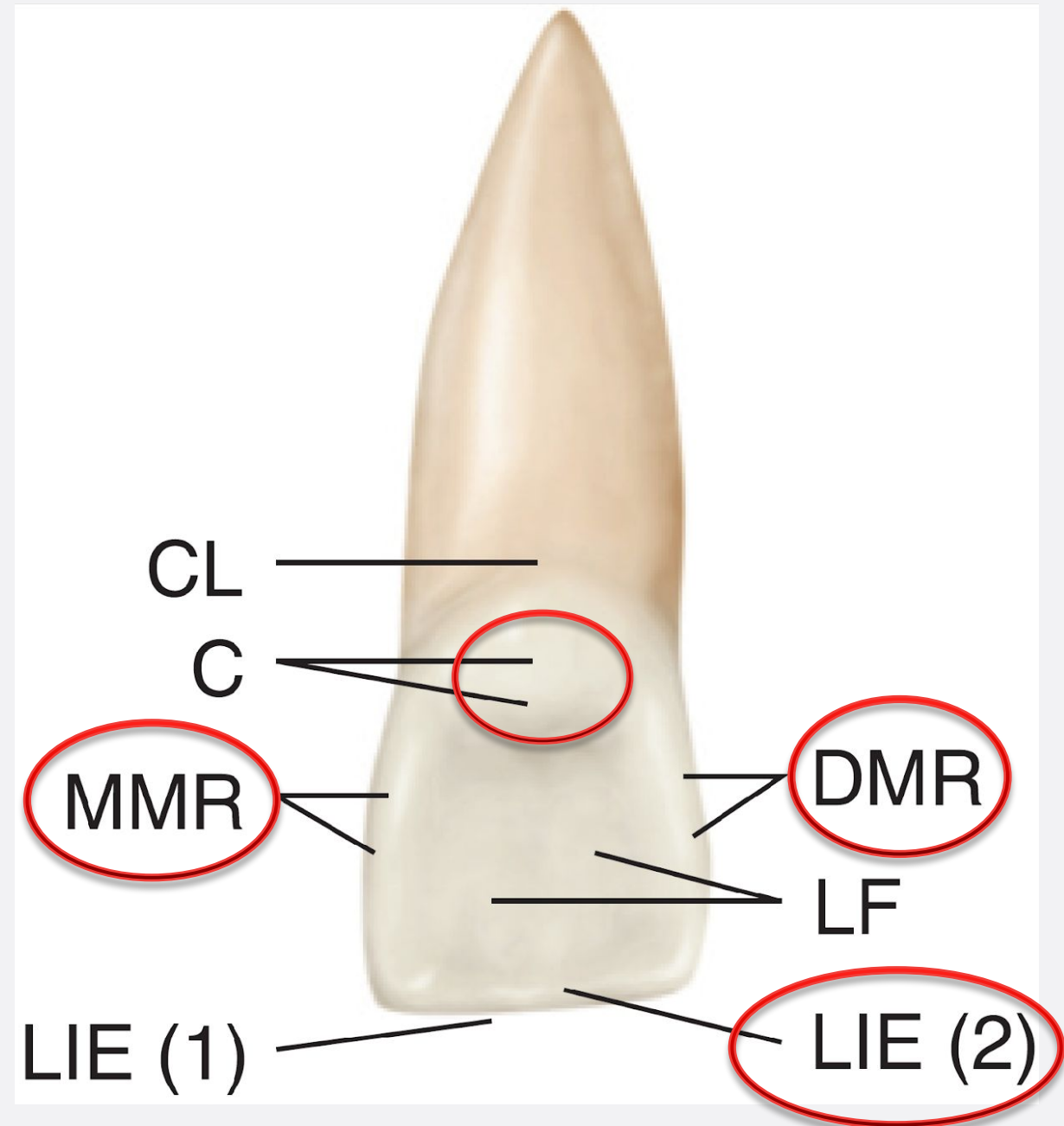
Lingual Surface

- A smooth convexity immediately below the cervical line = cingulum.
- Mesially and distally confluent with the cingulum = marginal ridges.
- A shallow concavity between the marginal ridges, below the cingulum = lingual fossa.

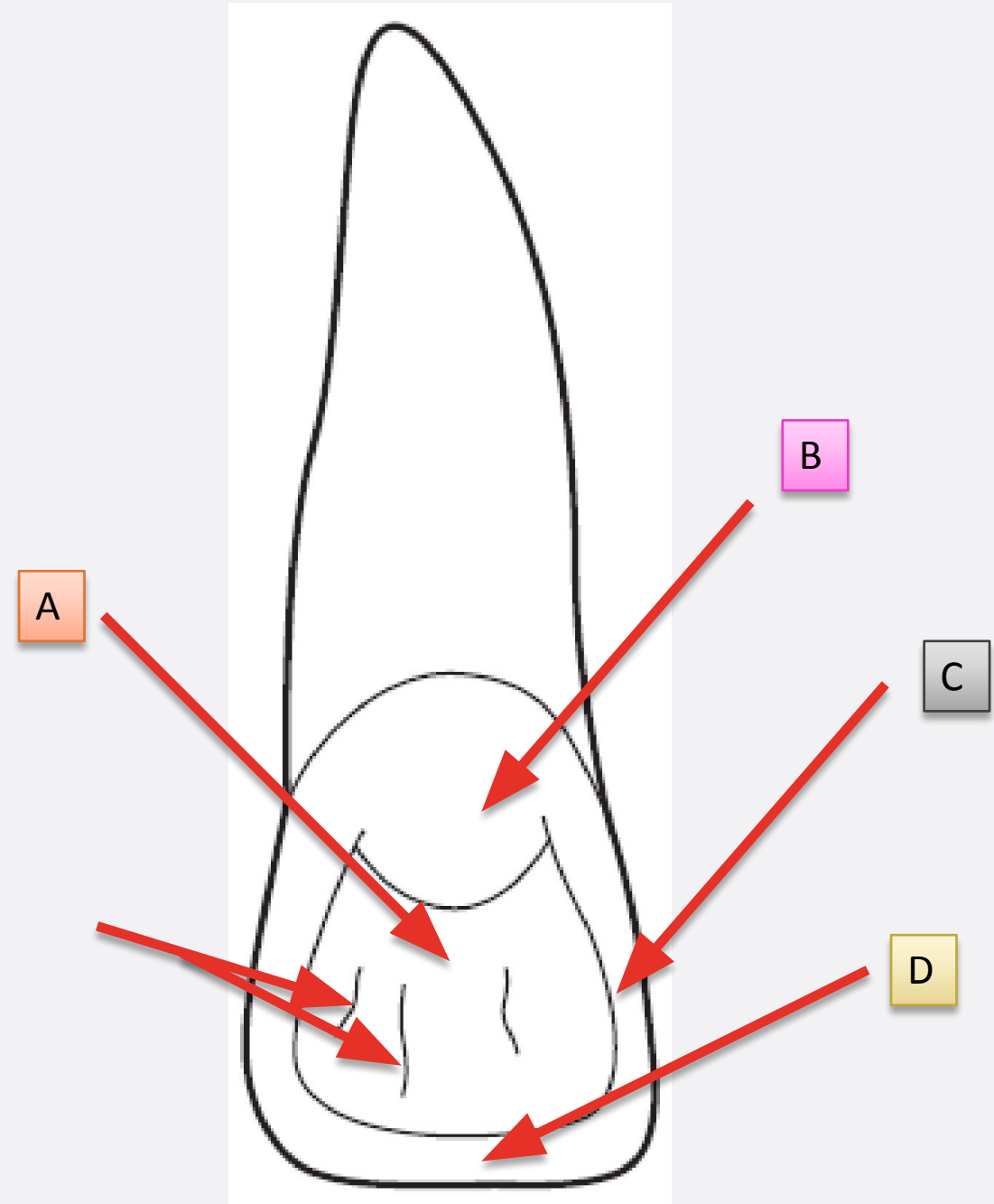


Lingual Surface

- **The lingual fossa** is bordered
 - ✓ **mesially** by the mesial marginal ridge,
 - ✓ **Incisally** by the lingual portion of the incisal ridge,
 - ✓ **distally** by the distal marginal ridge,
 - ✓ and **cervically** by cingulum

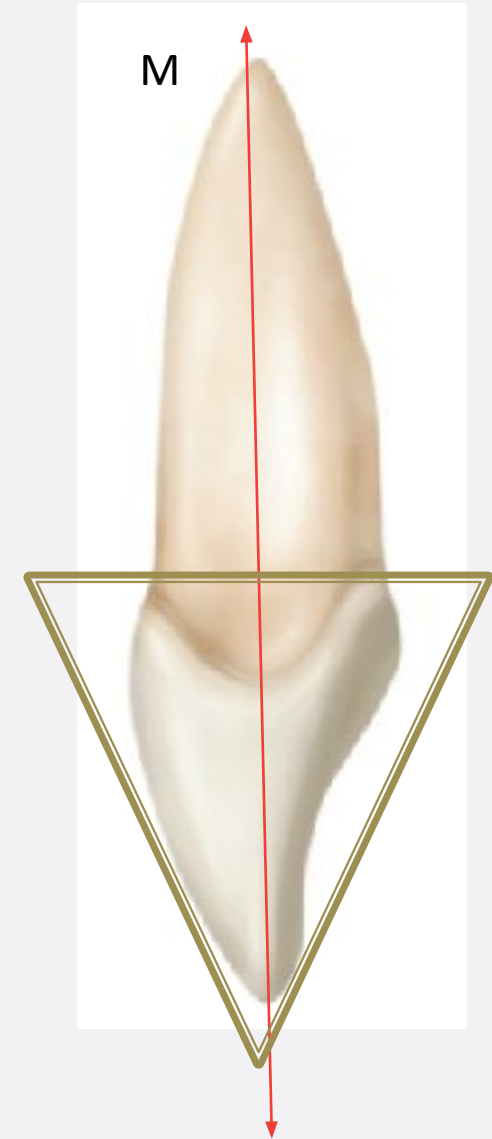


Identify the marked
feature on lingual
surface of the of
maxillary central
incisors!



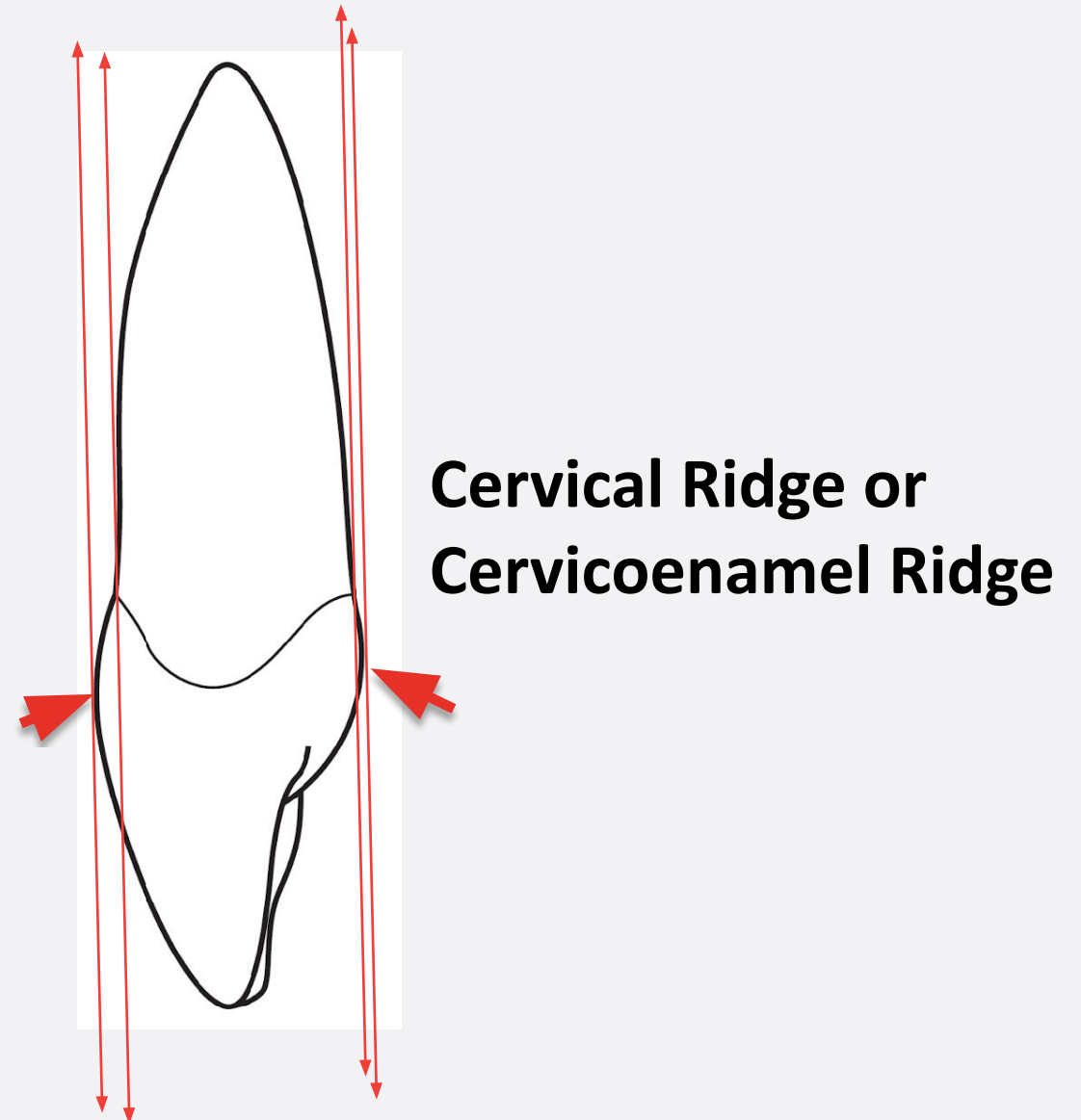
Mesial Surface

- The crown is wedge-shaped, or triangular, with the base of the triangle at the cervix and the apex at the incisal ridge.
- When a vertical line drawn through the crown and the root; incisal ridge of the crown lies on that line with the center of the root.



Mesial and Distal Surface

- Labially and lingually, immediately coronal to the cervical line are the crests of curvature of these surfaces = 0.5mm
- ✓ These crests of contour give the crown its greatest labiolingual measurement.



Mesial and Distal Surface

- The labial outline is very slightly convex.

- from the crest of curvature



- to the incisal ridge



D

Mesial and Distal Surface



M

- The lingual outline is convex at the crest of curvature at the cingulum;
- it then becomes concave at the mesial marginal ridge
- and slightly convex again at the linguo-incisal ridge and incisal edge.



D

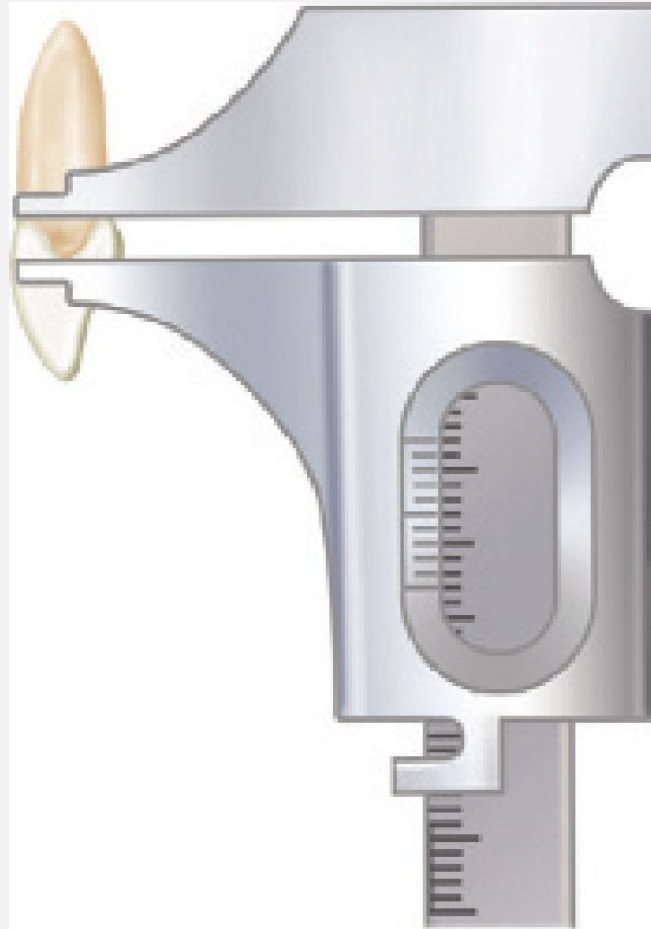
Distal Surface

crown gives the impression of being somewhat thicker toward the incisal third.



Dimensions of Maxillary Central Incisor

Curvature of Cervical Line— Mesial	Curvature Of Cervical Line— Distal
3.5mm	2.5mm



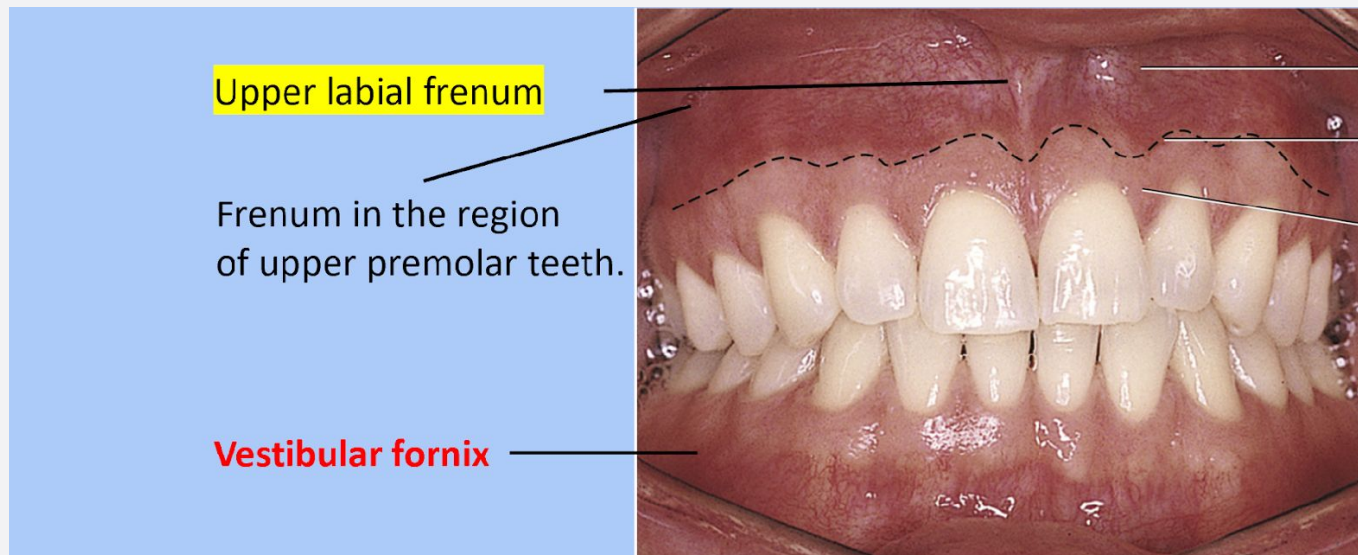
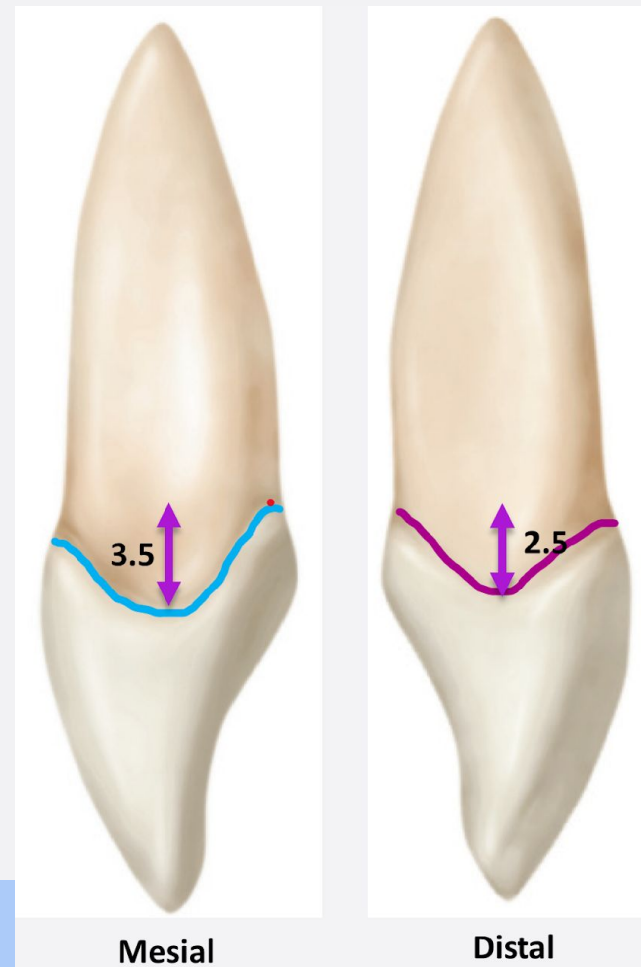
Mesial



Distal

Mesial and Distal Surface

- **Curvature of cervical line** represents the epithelial attachment of gingival tissue to the enamel of the crown (attached gingiva).
- It also outlines the CEJ.



Alveolar mucosa

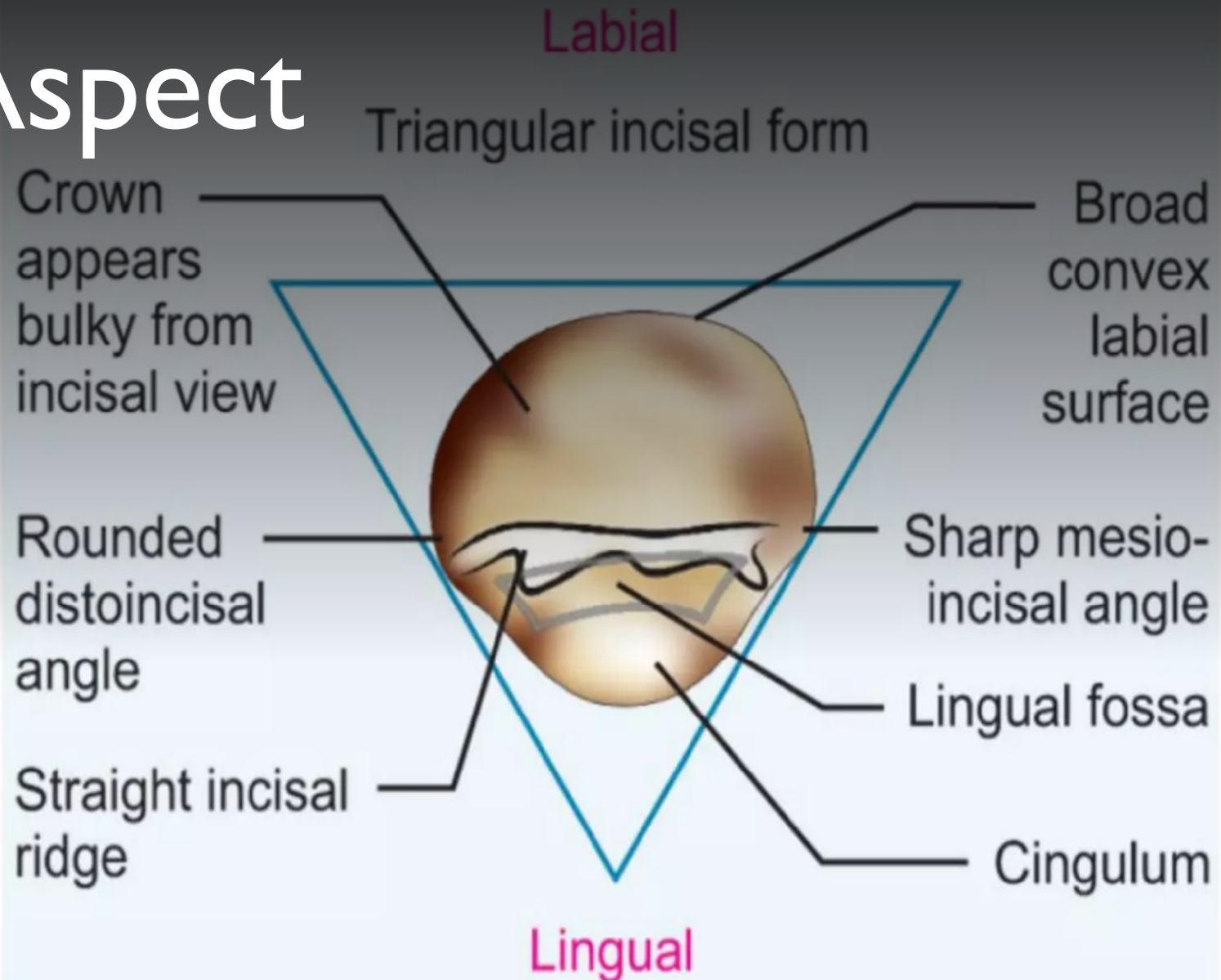
Mucogingival junction (Alveolar crest)

Attached gingiva

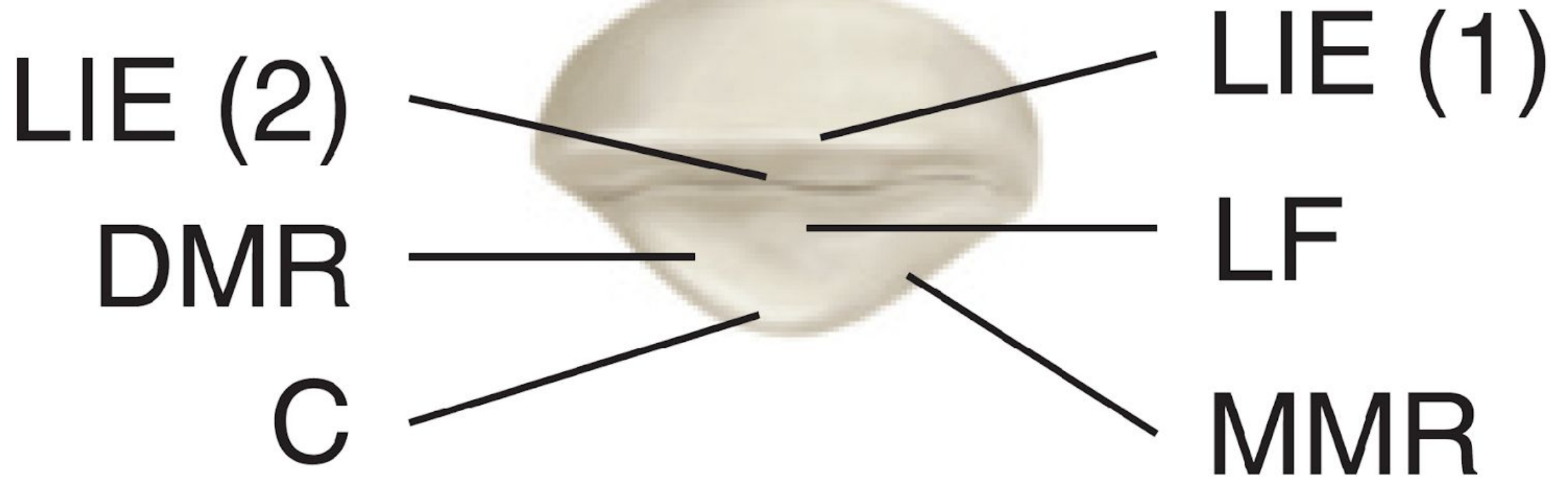
- **Patients came to dental OPD with complaint of mobility in maxillary central incisor. On clinical examination dentist found loss of epithelial attachment of gingival tissues around the tooth. There was complete loss of interdental papilla lying on the following surface of tooth.**

- a. Labial surface
- b. Lingual surface
- c. Proximal surface
- d. Incisal surface
- e. Mesial surface

Incisal Aspect



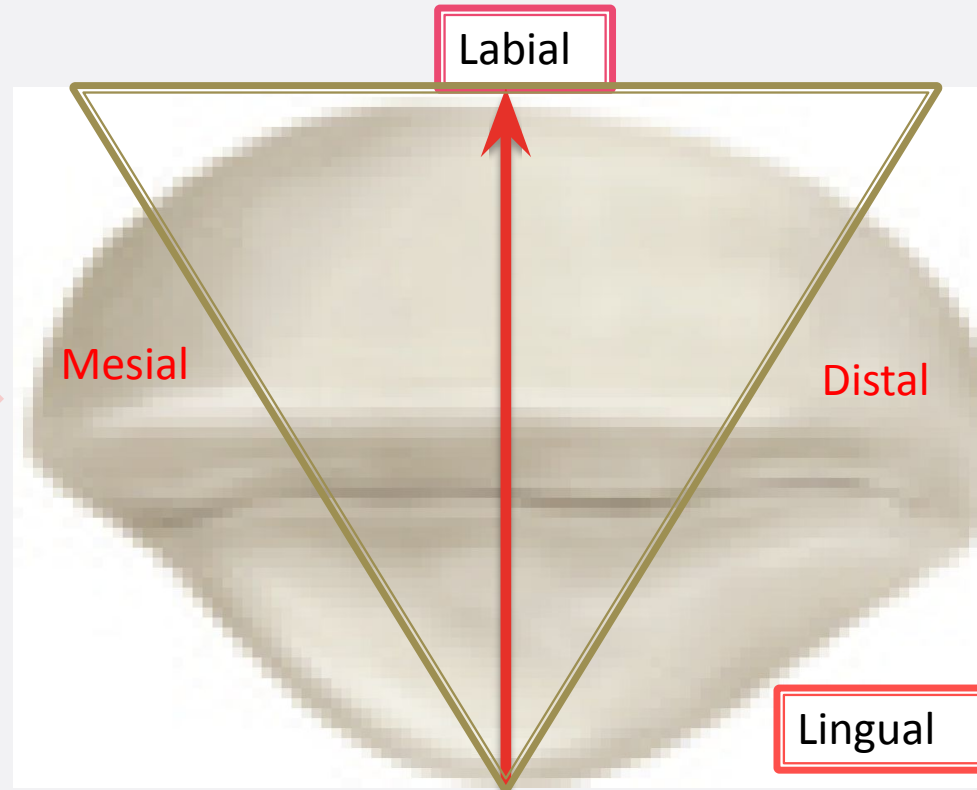
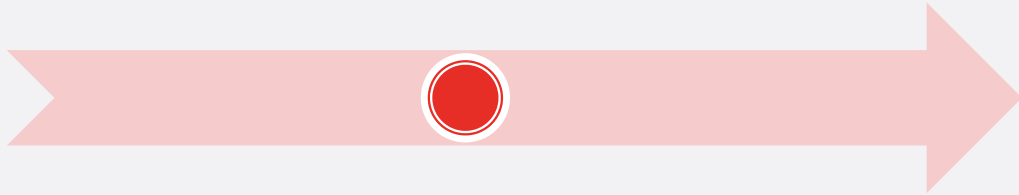
Incisal Aspect



Incisal Surface

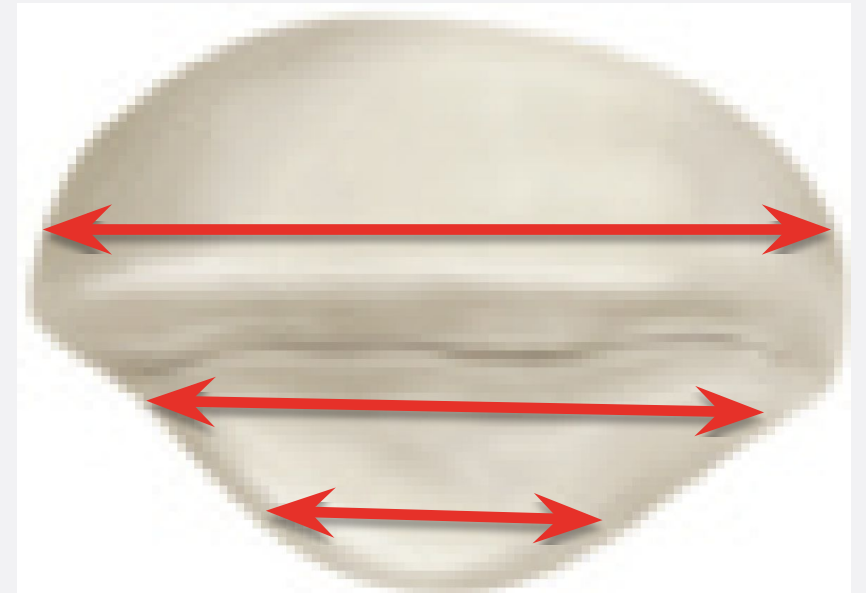
- Triangular cross section of the root at the cervix with rounded angles.
- One side of the triangle is labial, with the mesial and distal sides pointing lingually.

The mesial half of this triangle is slightly longer (more filled) than the distal side



Incisal Aspect

- **Broad convex labial surface compared to lingual surface, especially toward the incisal third.**



Incisal Surface

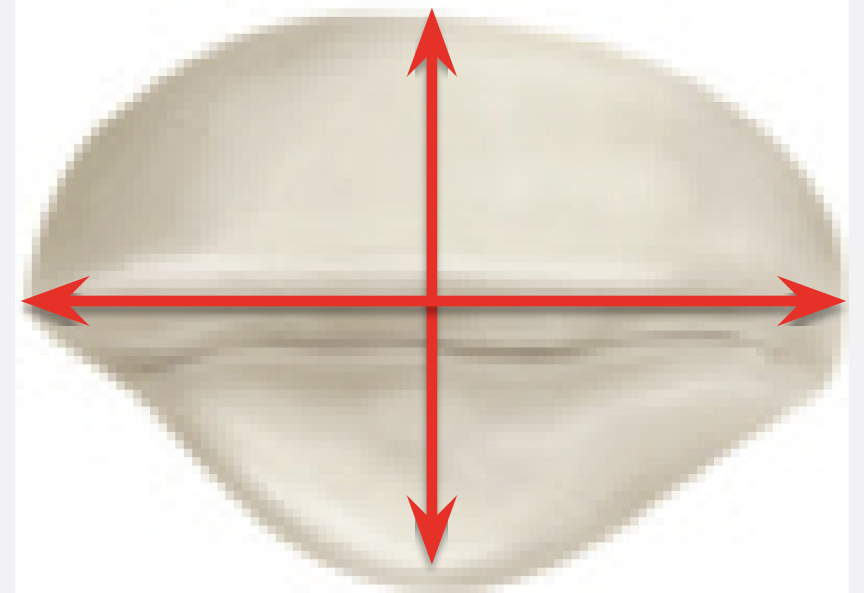
- The incisal ridge, slope lingually.
- Lingual portion tapers lingually toward the cingulum.



Incisal Surface

The labiolingual calibration of the crown is more than two-thirds as great as the mesiodistal calibration.

- Wider labiolingually than mesiodistally**



Labioincisal line angle

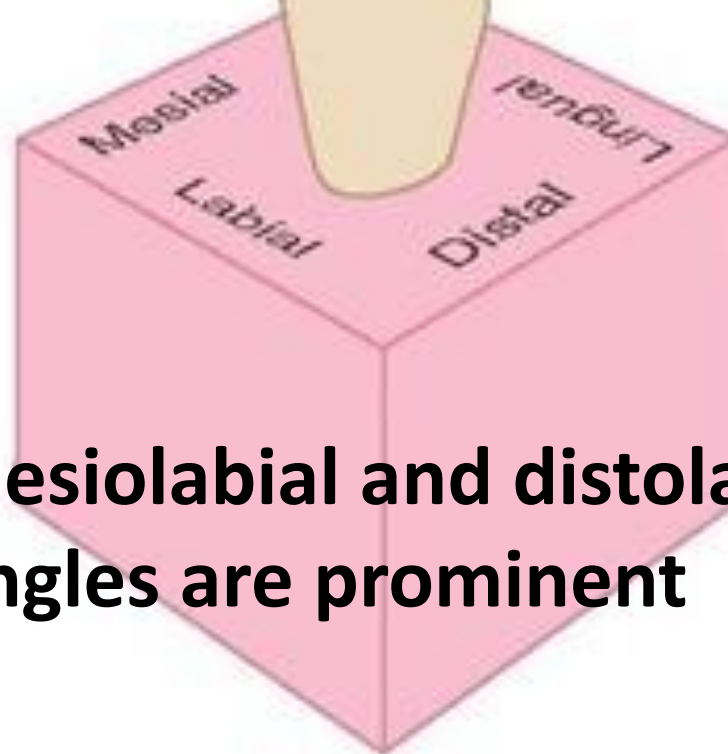
Mesiolingual line angle

Mesiolabial line angle

Linguoincisal line angle

Distolingual line angle

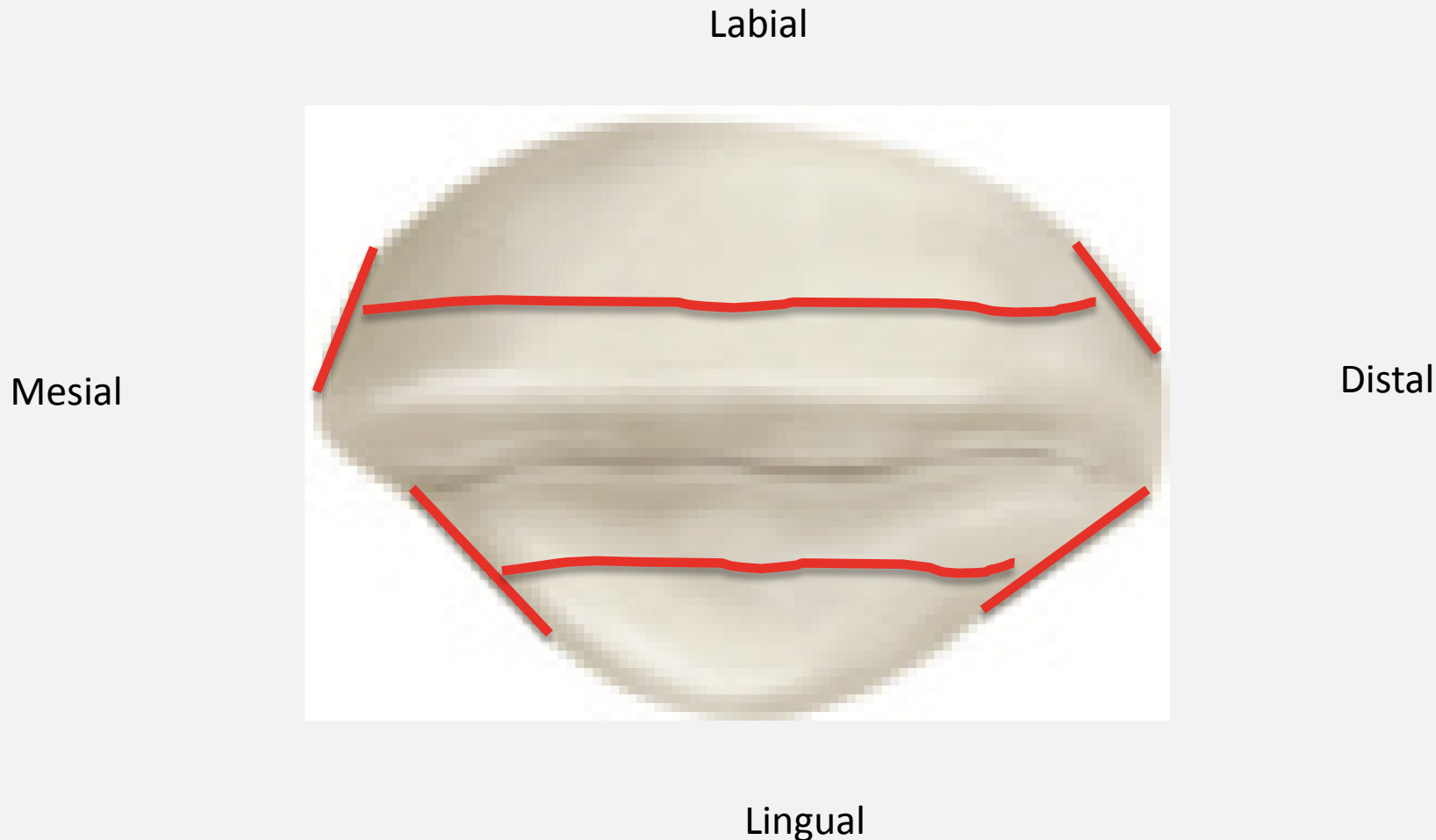
Distolabial line angle



The mesiolabial and distolabial line angles are prominent

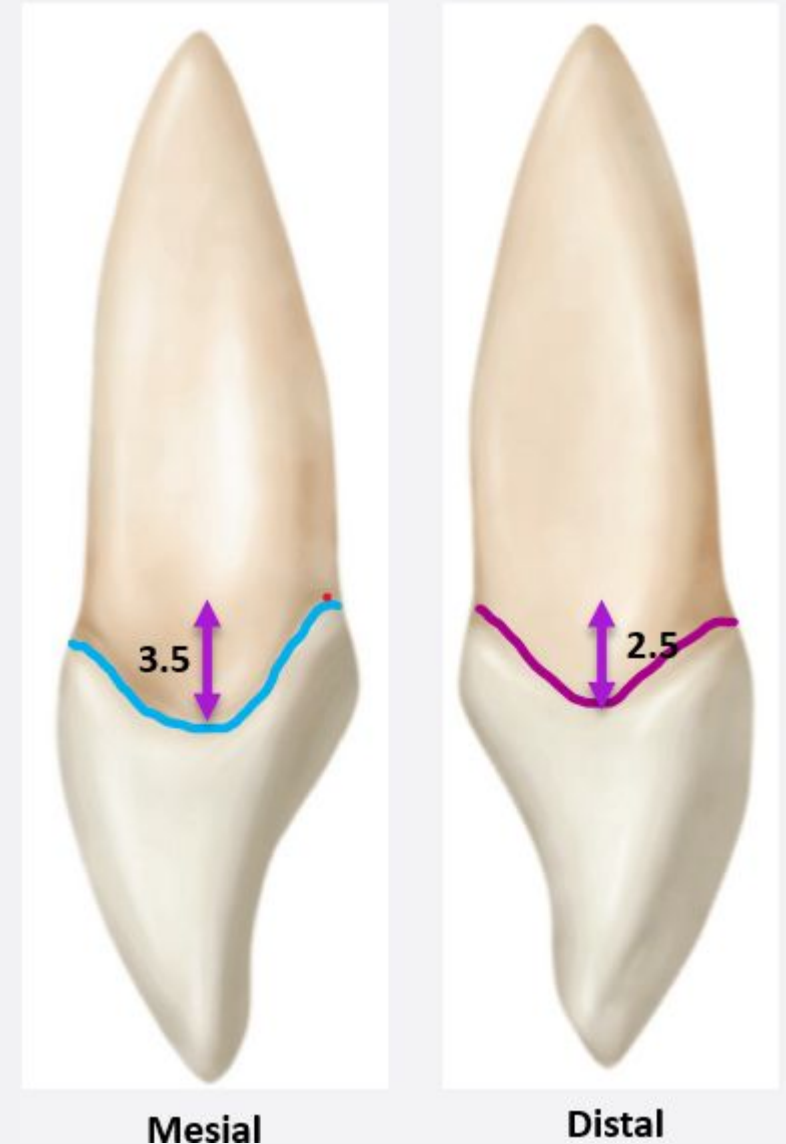


- The crown and root taper lingually, so that the crown calibration at the two labial line angles is greater than the calibration at the two lingual line angles, and the lingual portion of the root is narrower than the labial portion.



The Cervical Line - A Key Identifier

- The **mesial cervical line curvature** of the maxillary central incisor is **greater than on any surface of any other tooth** in the mouth.
- **Extent of Curvature:** ~3-4 mm incisally on a crown of average length.
- **Comparison:** The curvature on the distal surface is **less** than on the mesial.
- **Clinical Significance:** Important to recognize for:
 - **Scaling and Root Planing:** Avoid damaging the cementum.
 - **Crown Preparation:** Understanding the CEJ position.



Lingual Aspect - Clinical & Anthropological Facts

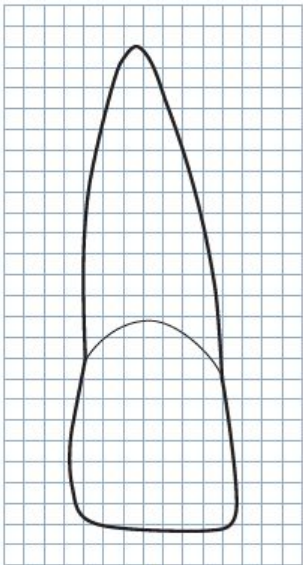
- **Variation: Shovel-Shaped Incisor**
 - An exaggeration of the marginal ridges.
 - Genetically determined trait.
 - Commonly seen in **Mongoloid races** (e.g., North and South American Indians, East Asian populations).
 - **Clinical Significance:** Can be mistaken for caries or affect orthodontic bracket bonding.



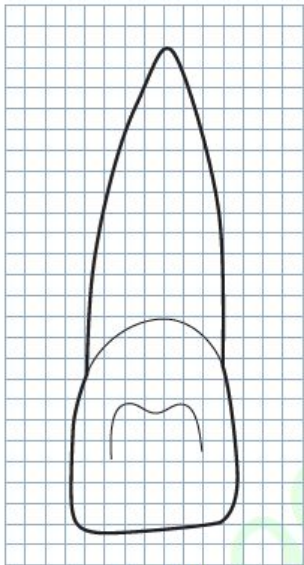
FIGURE 6-7
Maxillary right central incisor.
Squared millimeter graph
outlines of five aspects

	Length of Crown	Length of Root	Mesiodistal Diameter of Crown ^b	Mesiodistal Diameter of Crown at Cervix
Maxillary Teeth				
Central incisor	10.5	13.0	8.5	7.0

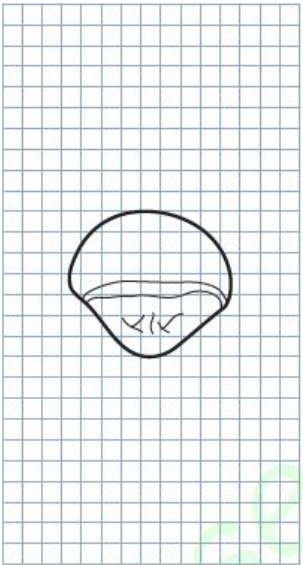
Labiolingual or Buccolingual Diameter of Crown	Labiolingual or Buccolingual Diameter of Crown at Cervix	Curvature of Cervical Line— Mesial	Curvature of Cervical Line— Distal
7.0	6.0	3.5	2.5



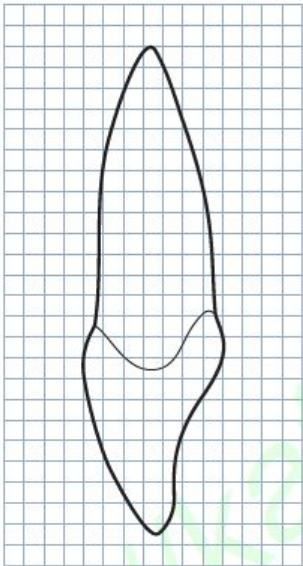
Labial



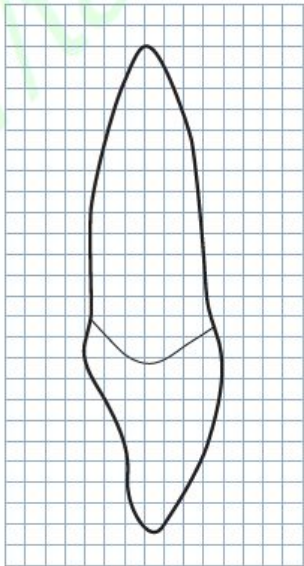
Lingual



Incisal



Mesial



Distal

Anatomical variations



- 1, Extralingual inclination of incisal portion of crown.**
- 2, Root extremely long. 3, small in all dimensions.**
- 4, Crown extremely long, root very short.**
- 5, Specimen malformed; crown unusually long; cervix very wide.**

Thank you!

